A Quantitative Convergence Law for the Connes–Consani–Moscovici Zeta Spectral Triple

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June 2026

Revised August 2026

Abstract

The Connes–Consani–Moscovici (CCM) construction produces finite real symmetric operators whose spectra approximate the ordinates of the nontrivial Riemann zeta zeros. We develop an operational convergence law for the first ordinate as a function of the prime cutoff λ^2, basis size N, and requested precision P. Intrinsically, the relative error separates into finite-basis and finite-cutoff channels,

$$E(N,\lambda^2) \le \varepsilon_{\text{modes}}(N,\lambda^2) + \varepsilon_{\text{primes}}(\lambda^2),$$

while the observed number of matching decimal digits is, up to a bounded noncancellation term,

$$D(\lambda^2,N,P) = \min\big(D_{\text{Wprec}}(P), D_{\text{nModes}}(N,\lambda^2), D_{\text{Primes}}(\lambda^2)\big) + O(1).$$

Precision is an instrument-resolution cap, not a third intrinsic error mechanism. On the calibrated domain $7 \le \lambda^2 \le 100$, the operational mode submodel is

$$D_{\text{nModes}} = D_{\text{Primes}} \tanh\left[\left(\frac{cN}{\lambda^2 \ln \lambda^2}\right)^{q_\gamma(\lambda^2)}\right], \qquad q_\gamma(\lambda^2) = \frac{2}{2\gamma + \ln\ln \lambda^2}, \qquad c = \frac{4}{\pi}.$$

With these published constants fixed, an independent numerical reimplementation reproduces complete ramps at $\lambda^2 = 7, 13, 20$ with RMS errors 0.47, 0.66, and 1.18 digits, respectively. A four-point sweep at $\lambda^2 = 1000$, independently reproduced by the Rust implementation, gives the complementary result: the same closed q_γ submodel overpredicts the first-root accuracy by 118–167 digits. Depending on the observable and fitting convention, the measured effective exponent lies in the range 0.74–0.80, rather than $q_\gamma(1000) = 0.648$. Thus the three-budget law, the stretched-tanh ramp family, and the coordinate $\lambda^2 \ln \lambda^2$ remain the working structure, while the large-cutoff dependence of the exponent, onset coefficient, or transition coordinate is explicitly open.

The finite-cutoff channel has the measured form $D_{\text{Primes}} = K\lambda^2 - P_m \log_{10} \lambda^2 - c_m$. Its leading coefficient $K = 4\pi/\ln 10$ coincides with the classical fourth-index prolate singular-value asymptotic; transfer to the constrained CCM h_0–h_4 trial state and the reverse inequality remain open. The retained lattice has the exact edge $T_{\max} = 2\pi N/\ln \lambda^2$. Defining the exact geometric capacity by $K_{\max}^{\text{geom}} = N_\zeta(T_{\max})$, the smooth Riemann–von Mangoldt approximation is

$$K_{\max}^{(0)}(N,\lambda^2) = \frac{N}{\ln \lambda^2}\left[\ln\left(\frac{N}{\ln \lambda^2}\right) - 1\right] + \frac{7}{8}.$$

This is a geometric reach statement, not an accuracy or ordering theorem. The empirical law is stated only for validated matrix assemblies. Explicit rigor tiers separate proved identities and bounds, conditional root-transfer formulas, calibrated laws, and open large-cutoff models. This work has been shared with the CCM authors, including their lead author, Professor Alain Connes.

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1 Introduction

The Connes–Consani–Moscovici (CCM) construction [1], building on the Connes–van Suijlekom truncated Weil form [2], produces a family of finite self-adjoint operators $D_{\log}^{(\lambda,N)}$ whose spectra approximate the imaginary parts γ_k of the nontrivial Riemann zeta zeros to remarkable numerical accuracy: 55 digits at the headline configuration of [1], which we extended to 1019.0 measured matching digits in an independent reproduction [51]. The construction has two integer-like control parameters, a prime cutoff λ^2 (the primes $p \le \lambda^2$ enter the Weil form) and a basis size N (the matrix is $(2N{+}1) \times (2N{+}1)$), together with the requested precision $P = P_{\mathrm{req}}$ in decimal digits. The rate at which the eigenvalue converges in the basis size N is named the central open problem in [1]; Groskin [7] measured a Galerkin convergence exponent empirically but disclaims deriving it, and Śliwiński [15] proves an inverse-logarithmic lower bound on the mean spectral error. We previously tested two proposed rate laws against high-precision data and found them unsupported [52].

The author's starting point is a belief the Hilbert–Pólya conjecture already expresses: that apparent randomness in the zeros is structure not yet written down. The CCM construction gives that belief a computable form. What remained open is the rate at which the computation converges, and characterizing that rate explicitly is the subject of this paper.

Definition 1. $D(\lambda^2, N, P) := -\log_{10}\big(|\nu_1 - \gamma_1|/|\gamma_1|\big)$ *is the* matching-digit count*: the number of correct leading decimal digits of the first Riemann zero γ_1 reproduced by ν_1, the first ordinate produced by the CCM construction, computed from the requested decimal target $P = P_{\mathrm{req}}$ via MPFR/GMP. The reference γ_1 is computed to* 2500 *significant digits by rigorous interval arithmetic (Arb [53]). The extended archive used in this paper reaches* ≈ 1070 *digits at* $\lambda^2 = 200$*, still far below the reference depth, so the reported counts are genuine and not reference-limited; its leading digits agree with the standard tabulations of [54].*

Notation (each object is defined where cited; collected here for reference)	
$E = \lvert\nu_1 - \gamma_1\rvert/\lvert\gamma_1\rvert$	relative error of the first ordinate; $D = -\log_{10} E$ (Def. 1)
$P = P_{\mathrm{req}}$	requested decimal target used by the operational law
P_{work}	internal decimal working precision after the guard reserve
$\varepsilon_{\mathrm{modes}},\ \varepsilon_{\mathrm{primes}}$	intrinsic error channels, eq. (3)
$D_{\mathrm{Wprec}},\ D_{\mathrm{nModes}},\ D_{\mathrm{Primes}}$	per-source digit budgets (Sections 4–6)
$\varepsilon^{\infty}(\lambda)$	smallest eigenvalue of the continuum form at cutoff λ
$D_{\mathrm{eig}} = -\log_{10}\varepsilon^{\infty}$	eigenvalue floor (Section 6)
$D_{\mathrm{edge}} = D_{\mathrm{Primes}} \cdot g$	the single edge function (Section 3.1)

The same number is read as a *ceiling* or a *floor* depending on the direction of view. D_{Primes} is a ceiling on the achievable digit count and, equally, the depth of the accuracy floor at the cutoff; both usages refer to the one object.

Our central result, the *Andrews CCM Convergence Law*, has two equivalent forms. Its intrinsic (spectral) form is the two-channel decomposition

$$\boxed{E(N, \lambda^2) \le \varepsilon_{\mathrm{modes}}(N, \lambda^2) + \varepsilon_{\mathrm{primes}}(\lambda^2)} \tag{1}$$

with the channels defined in Section 3, (3). Their signed displacements add exactly before absolute values are taken; Prop. 1 records the unconditional one-sided bound and the quantitative condition needed to exclude cancellation. The two channels are modeled as the finite-N transition and extreme tail of a proposed common prolate-type spectral edge (Section 3.1); convergence $\nu_1 \to \gamma_1$ is exactly $E \to 0$. Its finite-precision measurement is the matching-digit count

$$\boxed{D(\lambda^2, N, P) = \min\big(D_{\mathrm{Wprec}}(P),\ D_{\mathrm{nModes}}(N, \lambda^2),\ D_{\mathrm{Primes}}(\lambda^2)\big) + O(1)} \tag{2}$$

under the noncancellation condition of Prop. 1, and with an unconditional lower bound in its absence. Here D_{Wprec} is the instrument resolution alone. Since D is the relative error on a logarithmic scale, the law is a quantitative convergence statement: $D \to \infty$ is precisely $\nu_1 \to \gamma_1$, and the three budgets are the independent mechanisms that drive (or stall) that convergence (Prop. 7). Six points distinguish it from prior empirical fits:

1. **The minimum is forced, not fitted** (Section 3). In digit space D is a negative logarithm of an error bounded by a sum of operationally independent channels, so the largest binds and D is at least the minimum of the three per-source budgets, unconditionally, and equals it up to an $O(1)$ term absent cancellation (Prop. 1). This is the structural reason earlier single-formula digit fits failed. The precision budget is measurement, not mathematics: removing it leaves the single intrinsic edge (1), whose two channels (modes, primes) are the only mathematical content.

2. **Each budget is analyzed from the construction**, not guessed. The precision budget follows from elementary round-off analysis; the measured prime ceiling is compared with the fourth-index prolate asymptotic, with the CCM-to-prolate transfer stated explicitly as open; and the mode budget is related to the analyticity of the Weil-form eigenvector.

3. **Every claim carries an explicit rigor level** (Table 4), separating the rigorously proved parts from those derived and validated against high-precision computation. We do not label the whole law a theorem; we state precisely which pieces are.

4. **It separates fixed-cutoff convergence from the growing-cutoff rate** (Prop. 7). Under explicit simple/isolated/even hypotheses, the finite zero branch converges to a continuum-transform zero at fixed λ^2; if that limit differs from γ_1, increasing N and P cannot make that branch converge to γ_1. The proposed growing-cutoff rate is a separate, explicitly conditional statement.

5. **It identifies the finite spectral reach of the truncation** (Section 8.2). The retained scaling lattice has the exact outer frequency $T_{\max} = 2\pi N/\ln\lambda^2$. Combining this edge with the Riemann–von Mangoldt law gives a smooth estimate $K^{(0)}_{\max}$ for the number of zeta ordinates geometrically contained in the finite band. This is a capacity estimate, not a guarantee of accurate or correctly ordered recovery.

6. **No prior unified law, to our knowledge.** Earlier attempts are single closed-form fits of the digit count, and they fail: we tested two proposed rate laws against high-precision data and found them unsupported [52]. Groskin [7] likewise reports that no single smooth-convergence law fits his data and declines to assert a rate, and Śliwiński [15] establishes only a lower bound on the mean error. The unified characterization given here, the intrinsic two-channel edge (1) and its finite-precision measurement (2), is, to the best of the author's knowledge, the first working closed-form convergence law for the construction. Its unity is the point: the data and classical asymptotics support a common spectral-edge hypothesis (Section 7) for the mode ramp and prime ceiling, which offers a mechanism for why a single expression can hold where earlier per-fits could not.

The empirical claim has three distinct levels. First, the minimum-budget structure is the exact bookkeeping skeleton. Second, the stretched-tanh family describes the observed mode ramps. Third, the particular closed map $q_\gamma(\lambda^2) = 2/(2\gamma + \ln\ln\lambda^2)$ together with $c = 4/\pi$ is a fixed operational submodel. The first two levels survive every validation reported here. The third is accurate on the calibrated domain but is now known not to extrapolate unchanged to $\lambda^2 = 1000$. Stating those levels separately is essential: a measured failure of the large-cutoff parameter map is a constraint on the formula, not a retraction of the three-budget law or of the stretched-transition family.

The deepest new content is the mode budget D_{nModes} (Section 5): the measured matching-digit ramp is the logarithm of a stretched-exponential convergence, and its exponent is modeled by a sharp coefficient-decay law for the eigenvector (Gevrey where $q \leq 1$). Proving that regularity and its transfer to the root is an explicit research program, not an end-to-end derivation. A data-coordinate comparison explains how an eigenvalue-space local exponent of the kind measured by Groskin [7] can drift across a stretched-exponential window; equality of the two asymptotic observables remains conditional. Section 7 separates the rigorous Riemann–von Mangoldt counting backbone from the stronger conjecture that the mode ramp and prime ceiling are two scales of one CCM spectral edge.

2 The construction and the object D

We recall the construction of [1, 2]. Let $\lambda > 1$ and $L = 2\ln\lambda = \ln\lambda^2$. On $\mathcal{H}_\lambda = L^2([\lambda^{-1}, \lambda], du/u)$ the functions $V_n(u) = L^{-1/2} e^{2\pi i n \ln(\lambda u)/L}$ form an orthonormal basis. With $t = \ln u \in [-L/2, L/2]$, this centered convention is just the translation of the CCM basis $U_n(x)$ on $[0, L]$: $V_n(t) = U_n(t + L/2)$. The factor $(-1)^n$ that appears when V_n is written as an exponential in the centered variable is therefore a translation phase, not a different basis or a matrix twist. The Weil quadratic form QW_λ, restricted to $\mathrm{span}\{V_n : |n| \leq N\}$, is a real symmetric $(2N{+}1) \times (2N{+}1)$ matrix whose archimedean part carries the local factor at infinity and whose arithmetic part is a finite sum over prime powers $p^j \leq \lambda^2$. Reflection symmetry splits the matrix exactly into even and odd sectors. When the lowest even eigenvalue is simple and lies below the lowest odd eigenvalue, the global ground state ξ is even; this is the critical finite-section hypothesis in the CCM construction, not an identity valid without qualification. The positive zeros of the associated Fourier–Mellin function $\widehat{\xi}$ then approximate the ordinates γ_k [1].

The prime-power cutoff is likewise structural rather than a numerical quadrature convention. Compact logarithmic support exhausts the arithmetic translations beyond $p^j = \lambda^2$. As $u = \log\lambda^2$ varies, Groskin's finite-path analysis proves that the matrix is continuous and piecewise real analytic, with rank-one first-derivative jumps precisely at $u = \log(p^j)$ [9]. Thus the arithmetic cutoff, basis cutoff N, working precision, and any archimedean quadrature cutoff are four distinct controls. Groskin's finite Guinand–Weil dictionary gives an exact coefficient-vector-to-test-function translation. His Theorem 2.5 identifies every cutoff-free finite quadratic value with an absolutely convergent sum of an induced band-limited test function over the nontrivial zeros, without assuming RH [8]. His Theorem 3.2 and Corollary 3.3 also show that the omitted post-band archimedean tail is positive and provide an explicit one-sided decision budget. These results supply an independent audit of finite-cutoff assembly; they do not by themselves prove convergence of the selected CCM zero.

Two structural levels must be kept separate. At finite N, the closed form of $\widehat{\xi}_N$ is [1, Prop. 5.9], and under the simple-even ground state hypothesis [1, Thm. 5.10] proves that all its zeros are real and coincide with the spectrum of the perturbed finite operator. The general divided-difference mechanism is developed in [2, Sec. 5]. At the three finite configurations $(\lambda^2, N) = (13, 120), (29, 160), (53, 200)$, computer-assisted interval proofs certify that the lowest eigenvalue is simple and lies strictly in the even sector [13].[1] Combined with the finite CCM reality theorem, this certifies reality of every zero of the three associated finite transforms. The statement is finite-dimensional; it neither establishes a continuum sector gap nor proves growing-cutoff convergence.

At fixed cutoff and $N \to \infty$, [2, Thm. 6.1] gives compact-uniform transform convergence

[1] For the present audit, the certified matrices were checked entrywise against the CCM assembly: the pole block agrees algebraically with [1, Lem. 4.1], and the prime and archimedean sign conventions agree with [1, Props. 4.2–4.3]. The certificates are invariant under an additive diagonal-constant convention shift, because the matrix and the separating threshold shift by the same scalar.

and the continuum reality statement under a simple, isolated, even continuum ground state. Suzuki proves a positive, simple, even continuum ground state for sufficiently small aperture [5, Thm. 1.4] and gives a continuum reality theorem for the associated self-adjoint extensions [5, Thm. 1.5]. None of these results proves the growing-cutoff convergence of the selected zeros to the Riemann ordinates.

Appendix A rederives the finite transform formula and records the self-contained CCM proof chain. The pre-CvS foundation remains Weil's positivity criterion, Yoshida's small-support positivity theorem [16], Bombieri's truncated quadratic form [17], the archimedean operator theory of Connes–Consani [18], and Suzuki's screw-function framework [20, 21, 5].

Beyond the three certified finite configurations above, the selected eigenvector ξ is even and the smallest eigenvalue is simple throughout the reported above-floor numerical archive, with even symmetry verified across 38 configurations spanning $\lambda^2 = 13$–1000 in [51].

The object of study, $D(\lambda^2, N, P)$, is a digit count: $D = -\log_{10}(\text{relative error})$. The relative error itself, $E = |\nu_1 - \gamma_1|/|\gamma_1|$, is the intrinsic object of the law (1); D is its measurement in digits, and the arithmetic of that measurement is what makes the minimum structure of (2) inevitable.

3 The three-budget skeleton

The computed approximation ν_1 carries error from three sources: finite working precision (round-off), finite basis size (mode truncation), and the displacement of the fixed-cutoff continuum limit. We now *define* the channels, rather than posit their additivity. At a cutoff satisfying the spectral hypotheses of Section 8.3, write $\nu_1(N, \lambda^2)$ for the selected positive zero branch of the exact (infinite-precision) finite-N operator and $\nu_1(\infty, \lambda^2)$ for its $N \to \infty$ limit, and set

$$\varepsilon_{\text{modes}}(N, \lambda^2) := \frac{|\nu_1(N, \lambda^2) - \nu_1(\infty, \lambda^2)|}{|\gamma_1|}, \qquad \varepsilon_{\text{primes}}(\lambda^2) := \frac{|\nu_1(\infty, \lambda^2) - \gamma_1|}{|\gamma_1|}, \tag{3}$$

the finite-N displacement at fixed cutoff and the cutoff displacement at infinite basis size, with $\varepsilon_{\text{round}}(P)$ the round-off perturbation of the computed ν_1 relative to the exact finite-N value. The triangle inequality then gives, for the computed relative error,

$$E \;\le\; \varepsilon_{\text{round}}(P) + \varepsilon_{\text{modes}}(N, \lambda^2) + \varepsilon_{\text{primes}}(\lambda^2),$$

an inequality, not an identity: the signed displacements can in principle cancel. The channels are *independent* in a precise operational (not probabilistic or structural) sense: each answers to its own parameter (round-off to P, truncation to N, prime content to λ^2), so any one can be driven below the others at will. (Structurally they are coupled: $\varepsilon_{\text{modes}}$ depends on λ^2 through both the exponent q and the Shannon coordinate of Section 5, and below the precision floor round-off corrupts the eigenvector nonlinearly, Section 4.1; the operational sense is all Prop. 1 uses.)

Proposition 1 (The minimum is forced)**.** *Let $\varepsilon_1, \varepsilon_2, \varepsilon_3 \ge 0$ and let $E \ge 0$ satisfy $E \le \Sigma := \sum_i \varepsilon_i$. Then, unconditionally,*

$$D := -\log_{10} E \;\ge\; -\log_{10} \Sigma \;\ge\; \min_i\big(-\log_{10} \varepsilon_i\big) - \log_{10} 3,$$

since $\Sigma \le 3 \max_i \varepsilon_i$. If in addition no cancellation occurs — quantitatively, $E \ge \theta \max_i \varepsilon_i$ for some fixed $\theta \in (0, 1]$ — then also $D \le \min_i(-\log_{10} \varepsilon_i) + \log_{10}(1/\theta)$, whence $D = \min(D_{\text{Wprec}}, D_{\text{nModes}}, D_{\text{Primes}}) + O(1)$.

The lower bound, that the computed digit count is at least the binding budget minus $\log_{10} 3 < 0.48$ of a digit, is a rigorous consequence of the triangle inequality alone. The matching upper bound requires the explicit noncancellation condition $E \ge \theta \max_i \varepsilon_i$. Its additive

constant is $\log_{10}(1/\theta)$; it is not bounded by $\log_{10} 3$ without an independent lower bound on θ. Cancellation is most plausible near crossover seams, where two channels have comparable magnitude, but the triangle inequality alone neither confines it to those seams nor excludes an isolated cancellation elsewhere. Empirically, no seam in the recorded starvation sweeps shows D exceeding the binding minimum by more than the fitted bounded offset. Accordingly, the equality form used below is a quantitatively testable noncancellation model, while the one-sided bound is unconditional. Its content is that the *binding mechanism* changes across the parameter space (round-off, mode truncation, then prime content), not that D is literally nonsmooth (the negative logarithm of a signed near-sum can be smooth and can also exhibit cancellation spikes).

This is why, in digit space, D is a minimum rather than a sum or product, and why global single-expression fits of the digit count fail in practice: a single elementary term cannot simultaneously hold a hard precision wall, a prime ceiling, and a λ^2-coupled mode ramp. In the intrinsic form the error is, by contrast, the two-channel decomposition (1); the digit-space minimum is what the negative logarithm of that near-sum becomes, the largest term binding. The junctions differ in character. The precision junction is a hard wall, since round-off is a clamp, so its corner is sharp. The mode→prime junction, by contrast, is observed as a wide, smooth saturation (the tanh of Section 5). The measured digit count approaches the fixed-cutoff limit as $N \to \infty$ along the selected branch. Neither a pointwise inequality $D_{\mathrm{nModes}} \le D_{\mathrm{Primes}}$ nor monotone root accuracy is asserted: finite-N displacement can partially cancel the cutoff displacement.

The starvation protocol. Each budget is isolated experimentally by *starvation*: to read D_i directly, set the other two parameters large enough that their budgets exceed D_i, so the channel of interest dominates E outright and the cancellation caveat of Prop. 1 is vacuous. This protocol is the experimental mirror of the minimum structure. It is what makes the digit form falsifiable rather than descriptive, since each budget's functional form is measured in a regime where it alone binds. We use it throughout Sections 4–6.

Validity premise for empirical calibration. The three budgets begin after the finite CCM matrix has been assembled correctly. Matrix-assembly error is not a fourth intrinsic convergence mechanism and is not added to the operational law. It is instead an admissibility condition on calibration and validation data. Deep spectra can be stable under increased arithmetic precision while still being wrong because of an inadequate archimedean cutoff or a precision-stable special function defect. Accordingly, a state enters the empirical archive only when its assembly is supported by an independent cutoff or closed-form check, structural symmetry tests, residual verification, and recorded reference-zero provenance. This follows the verified-computation principle that numerical work should check the hypotheses of an analytic bound rather than infer correctness from a small residual alone [39]. In the present matrix family, Groskin's tail-order theorem proves that the omitted post-band archimedean increment is positive and decays only at order $\log T/T$; a negative finite-T eigenvalue inside the explicit tail budget certifies nothing about the cutoff-free form [8, Thm. 3.2 and Cor. 3.3]. The independent quadrature and closed-form audits [40, 41] make the same requirement concrete at implementation level. The assembled diagonal in the present implementation was also checked entrywise to 40 digits against the printed combined constant in [1, Sec. 4.3], independently of the quadrature route.

3.1 The two forms of the law

Of the three budgets, only two (mode truncation and prime cutoff) are properties of the finite operator; round-off is a property of the computation. Separating them gives the two forms stated in Section 1.

Intrinsic (spectral) form. The mathematical content is the two-channel decomposition (1), with

$\varepsilon_{\text{modes}} \sim \exp(-aN^q) \to 0$ as $N \to \infty$ (Section 5) and $\varepsilon_{\text{primes}} \to 0$ as $\lambda^2 \to \infty$ (Section 6). It makes no reference to precision. Under (H^*_λ), the mode displacement from the selected fixed-cutoff continuum zero vanishes as $N \to \infty$; the decay of the continuum-zero displacement as $\lambda^2 \to \infty$ is the separate growing-cutoff premise tested in Section 6. Vanishing of both channels is sufficient for $E \to 0$. The converse is not automatic for signed errors because finite-section and cutoff displacements may cancel. At a fixed cutoff satisfying (H^*_λ), however, the selected branch converges to $\nu_\infty(\lambda) \neq \gamma_1$, so increasing N alone cannot make that branch converge to γ_1 (Prop. 7).

Observed (digit) form. A computation with requested target $P = P_{\text{req}}$, realized internally at P_{work}, resolves E only down to its round-off floor, so the measured matching-digit count is

$$D(\lambda^2, N, P) = \min\big(D_{\text{Wprec}}(P),\, -\log_{10} E(N, \lambda^2)\big) = \min\big(D_{\text{Wprec}}, D_{\text{nModes}}, D_{\text{Primes}}\big) + O(1), \quad (4)$$

where the second equality is conditional on a parameter-uniform noncancellation bound as in Prop. 1. Here $D_{\text{Wprec}}(P)$ is the instrument resolution: it is no property of ν_1, γ_1, or the operator, and as $P \to \infty$ it disappears, returning the intrinsic form in digit units. Thus the boxed law (2) is the finite-precision *measurement* of the intrinsic law (1).

One-edge parametrization. The data admit a useful common-edge parametrization of the mode and prime channels. Dividing out the ceiling (Section 5) writes $D_{\text{nModes}} = D_{\text{Primes}} \cdot g$ with $g = \tanh\big((c\,N/(\lambda^2 \ln \lambda^2))^q\big) \uparrow 1$ as $N \to \infty$, so $D_{\text{Primes}} = \lim_{N\to\infty} D_{\text{nModes}}$ is the $N \to \infty$ ceiling of the mode ramp in this parametrization; equivalently the digit law can be written using $D_{\text{edge}}(N, \lambda^2) = D_{\text{Primes}} \cdot g$ capped by precision, $D = \min(D_{\text{Wprec}}, D_{\text{edge}}) + O(1)$. Algebraic success of this parametrization does not prove that both errors arise from one spectral object. The Landau–Widom transition and Fuchs tail provide the evidence for that stronger common-edge hypothesis, formulated with its required comparison theorems in Section 7.

4 The precision budget D_{Wprec}

The requested decimal target is $P = P_{\text{req}}$. The implementation converts that target to an internal precision P_{work} by allocating $\lceil P_{\text{req}} \ln 10 / \ln 2 \rceil + 64$ bits; the guard reserve contributes approximately 19.27 decimal digits before accumulation and conditioning losses. For fixed (λ^2, N), a backward-stable computation has unit round-off proportional to $10^{-P_{\text{work}}}$. Its forward root error is bounded by an effective condition factor κ_{eff} that includes matrix assembly, separation of the selected eigenpair, transform evaluation, and the inverse slope at the selected simple zero. If these factors and the algorithmic backward-error constant remain independent of the requested target, then

$$D_{\text{Wprec}}(P_{\text{req}}) = P_{\text{req}} - \log_{10} \kappa_{\text{eff}} + O(1) = P_{\text{req}} + O(1), \quad (5)$$

where the guard reserve is absorbed in the implementation-dependent constant. This is a fixed-problem stability statement, not a claim that the condition factor is uniform as N or λ^2 grows. At $\lambda^2 = 13$ and $N = 80$, the validation sweep $P \in \{30, 45, 90, 180\}$ gives $D \in \{49.238, 55.654, 55.654, 55.654\}$, exhibiting the switch from the working-precision budget to the fixed-cutoff prime ceiling. Isolating this budget at $\lambda^2 = 50$ and $N = 400$, a sweep over $P \in \{50, 100, 150, 200\}$ gives an endpoint slope of 0.99970 and an ordinary least-squares slope of 1.00205 with $R^2 = 0.99994$, consistent with the predicted unit precision slope. The exact offset is not a property of the construction and we do not assign it physical meaning. *Status: the unit slope is T1 conditional on a fixed, backward-stable problem with a simple selected root; backward stability of the complete implementation is supported by the precision sweep and independent replay checks, not proved here.*

4.1 Symmetry, positivity, and the precision floor

A question the precision floor settles, and one easy to answer wrongly without it, is whether the even symmetry of the ground eigenvector, and the positivity of ε_N, are genuine features of the operator or merely of the regime above the floor. The CCM construction [1] posits a simple smallest eigenvalue with an *even* eigenvector (its Step 1) and enforces this with a forced-even projection. The reproduction of [51] tested that projection directly: across 38 configurations ($\lambda^2 = 13$–1000, $N = 10$–800, spanning both sides of the precision floor) with the projection *disabled*, the natural smallest eigenvector converged to the same even state produced by the projection, with numerically equivalent eigenvalues, at every tested configuration above the floor. The forced-even projection is thus empirically a no-op over that tested range: even symmetry is exhibited by the unprojected finite operator there, not manufactured by the projection.

The qualifier "above the floor" is essential, and it is precisely where the budget of this section bites. When the target eigenvalue lies below D_{Wprec}, smaller than the round-off that the working precision can resolve, the computed eigenvector is dominated by numerical noise: it can present spurious mixed symmetry, and the smallest computed eigenvalue can carry a spurious sign. These are artifacts of precision starvation, not properties of the operator; raising P past the floor dissolves them, and the natural eigenvector is again even, into the deep near-zero spectrum. (This is also the regime in which the round-off channel of Section 3 stops acting as an additive perturbation of ν_1: below the floor it corrupts the eigenvector nonlinearly, so the decomposition (3), like the symmetry and positivity statements of this subsection, is an above-floor statement.)

This even, positive ground state was documented in our reproduction [51]. In subsequent correspondence, and in a revision of his own paper [7], Groskin independently corroborated it by a different route: the sign changes his separate quadrature implementation had shown among the deepest eigenvalues at a finite archimedean cutoff T were identified as a finite-T artifact that vanishes as T is increased, consistent with the naturally even, positive ground state of [51]. Working precision P and archimedean cutoff T are therefore independent controls: a deep-spectrum value is trustworthy only when it is stable across two settings of *each*, not when it merely holds steady in one. Across the tested and independently corroborated configurations, the even, positive ground state persists into the deep near-zero spectrum, while observed breakdowns disappear when the relevant precision or cutoff is raised. A uniform statement for the full growing-cutoff path remains the sector-gap problem of Section 10.

5 The mode budget D_{nModes}

This is the convergence rate in N, the open problem of [1]. In the intrinsic form it is the mode channel $\varepsilon_{\mathrm{modes}}$; as a digit count it is the rising transition in the proposed common-edge parametrization (Section 7), climbing toward the prime ceiling. Dividing out that ceiling (Section 6), write the ramp ratio $g = D_{\mathrm{nModes}}/D_{\mathrm{Primes}} \in [0,1]$. On the validated closed-submodel domain $7 \le \lambda^2 \le 100$ we use

$$\begin{aligned} D_{\mathrm{nModes}}(N,\lambda^2) &= D_{\mathrm{Primes}}(\lambda^2)\,\tanh\Big(\big(cN/(\lambda^2\ln\lambda^2)\big)^{q_\gamma(\lambda^2)}\Big), \\ q_\gamma(\lambda^2) &= \frac{2}{2\gamma+\ln\ln\lambda^2}, \qquad c=\frac{4}{\pi}, \qquad 7\le\lambda^2\le 100. \end{aligned} \tag{6}$$

The stretched-tanh family and the coordinate $\lambda^2\ln\lambda^2$ are tested more broadly; the statement above is the fixed operational parameter map, not a claim that q_γ and c persist unchanged at arbitrary cutoff. We build the model in four steps: the stretched-exponential meaning of the ramp (5.1), the candidate archimedean rate (5.2), the prime correction giving q (5.3), and the Shannon coordinate (5.4).

5.1 The ramp is a stretched exponential

Proposition 2 (Stretched-exponential reparametrization)**.** *Suppose the pre-saturation mode budget has the asymptotic form*

$$D_{\mathrm{nModes}}(N,\lambda^2) = \frac{a}{\ln 10} N^q + r(N), \qquad r(N) = o(N^q),$$

for some $a = a(\lambda^2) > 0$ and exponent $q = q(\lambda^2)$. Then:

(i) (Stretched-exponential.) *The finite-section root error obeys*

$$\varepsilon_{\mathrm{modes}}(N,\lambda^2) = \exp\bigl(-aN^q + o(N^q)\bigr).$$

(ii) (Local power-law coordinate.) *If r is differentiable as a function of $\ln N$ and $dr/d\ln N = o(N^q)$, then the logarithmic local power-law exponent satisfies*

$$2s_{\mathrm{loc}}(N) := -\frac{d\ln \varepsilon_{\mathrm{modes}}}{d\ln N} = a\,q\,N^q + o(N^q).$$

Proof. Part (i) follows from $\ln \varepsilon_{\mathrm{modes}} = -(\ln 10) D_{\mathrm{nModes}} = -aN^q + o(N^q)$. Differentiating this identity with respect to $\ln N$ and using the stated derivative control on r gives part (ii). □

The proposition is an exact change of coordinates for the measured root error; by itself it does not identify an eigenvector regularity class. The proposed mechanism is sharper: the same exponent is conjectured to govern the phase-aligned eigenvector tail. In standard terminology this is the *Gevrey scale*: the Gevrey class G^μ, $\mu \geq 1$, is characterized by Fourier-coefficient decay $\exp(-c\,|k|^{1/\mu})$, reducing to strip-analytic functions at $\mu = 1$ [30, 42]. Thus for $0 < q \leq 1$ the statement $|\xi_k| \sim \exp(-c\,k^q)$ is the sharp statement $\xi \in G^{1/q}$ (and no better). When a finite-range fit gives $q > 1$, as at $\lambda^2 = 7$ in Table 1, it instead describes an ultra-analytic coefficient-decay scale; calling that ordinary Gevrey regularity would be incorrect. Establishing the coefficient law, and then transferring it to the transform zero, are separate open steps. The open "variance-to-exponent map" of Section 5.3 is thereby a precisely posed regularity lemma on a classical scale: *prove the sharp coefficient-decay class with exponent $q(\lambda^2)$ and, where $q \leq 1$, membership in $G^{1/q(\lambda^2)}$, with $1/q = \gamma + \frac{1}{2}\ln\ln\lambda^2$.* The measured exponents cross the analytic value $q = 1$ near $\lambda^2 = 13$ and then decrease with λ^2 (Table 1). This is consistent with a transition from ultra-analytic finite-range decay at the smallest cutoff to a prime-roughened Gevrey scale, but that regularity interpretation remains conjectural.

The expanded eigenstate archive supplies a direct stress test of this interpretation. Applying one fixed fractional-window fit to the stored coefficient envelopes does not recover a cutoff-only exponent: at $\lambda^2 = 250$, for example, the effective exponent falls from about 1.72 at $N = 60$ to about 0.99 at $N = 1500$, and it also moves when the fitted part of the finite vector is changed. This does not refute a sharp law for the continuum eigenvector, because the available vectors are finite sections and their edges move with N. It does rule out treating the exponent fitted from the root-error ramp as an already measured coefficient exponent. A promotion of the conjecture must therefore prove a fitting-window-independent continuum tail statement and a separate tail-to-root transfer; finite coefficient plots alone are insufficient.

The force of part (i) is that it converts the open problem of *measuring* a convergence rate into the analytic question of *explaining* the observed exponent: the single exponent q fixes the entire pre-saturation curve. It also explains why a fixed power-law fit drifts across an actually stretched-exponential window and resists extrapolation (Section 9).

Part (ii) gives a direct data-level reconciliation with the local exponent $s(c)$ measured by Groskin [7], provided the two experiments are sampling the same asymptotic error channel; this comparison is taken up in Section 9. Separately, for a simple isolated eigenvalue the Rayleigh-quotient error is second order in the phase-aligned eigenvector error, $|\varepsilon - \varepsilon_N| = O(\|\xi - \xi_N\|^2)$

[37, 43, 44, 45]. That result concerns the Rayleigh eigenvalue. It does not by itself square the displacement of a zero of $\widehat{\xi}_N$; the latter requires the quantitative tail-to-transform estimates and the simple-zero perturbation mechanism stated below. *Status: the reparametrization and local-slope identities are T1; their eigenvector-regularity and cross-observable interpretations remain conditional.*

5.2 The candidate archimedean rate $\delta = \pi/4$

The exact archimedean input is the completed real Gamma factor $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$. Its logarithmic derivative $\psi(\frac{1}{4}+\frac{iz}{2}) \sim \log(z/2)$ supplies the large-frequency $\log|z|$ symbol identified by Suzuki [5], while Stirling gives the real-axis envelope $|\Gamma(\frac{1}{4}+\frac{iz}{2})| \asymp |z|^{-1/4}\exp(-\pi|z|/4)$. Thus the exact archimedean envelope rate is

$$\delta = \pi/4. \tag{7}$$

The same value follows differentially. On the critical line the archimedean log-derivative has the large-z expansion

$$\psi\big(\tfrac{1}{4}+\tfrac{iz}{2}\big) = \log\frac{z}{2} + i\,\frac{\pi}{2} + o(1), \qquad z \to +\infty, \tag{8}$$

since the argument $\frac{1}{4}+\frac{iz}{2}$ runs in the imaginary (critical-line) direction, so $\arg\big(\frac{1}{4}+\frac{iz}{2}\big) \to \pi/2$. Its real and imaginary parts carry complementary information:

- the *real* part $\operatorname{Re}\psi(\frac{1}{4}+\frac{iz}{2}) \to \log(z/2)$ is the large-frequency symbol, Suzuki's $\log|z|$ multiplier [5];
- the *imaginary* part $\operatorname{Im}\psi(\frac{1}{4}+\frac{iz}{2}) \to \frac{\pi}{2}$ sets the envelope, through

$$\frac{d}{dz}\log\big|\Gamma_{\mathbb{R}}(\tfrac{1}{2}+iz)\big| = -\tfrac{1}{2}\operatorname{Im}\psi\big(\tfrac{1}{4}+\tfrac{iz}{2}\big) \longrightarrow -\tfrac{1}{2}\cdot\tfrac{\pi}{2} = -\tfrac{\pi}{4}.$$

Both statements arise from the one Gamma factor. The elementary identity

$$\delta = \frac{\pi}{4} = \frac{1}{2}\cdot\frac{\pi}{2} = \tfrac{1}{2}\arg(i), \tag{9}$$

records the half-angle generated by the critical-line direction. It is an envelope identity, not a proof of an analyticity strip for the finite CCM ground state.

Two transfer steps remain open. First, the finite ground-state transform, which is of exponential type $L/2$, is not literally the completed xi function; one must prove that its phase-aligned coefficient tail inherits the $\Gamma_{\mathbb{R}}$ envelope in the relevant growing window. Second, one must prove how that tail controls the selected transform-zero displacement, including whether the observed exponent is doubled. The measured finite coefficient rate (≈ 0.71 at $\lambda^2 = 100$) is therefore a separate, prime-modulated quantity rather than an instance of a proved $\pi/4$ law. The exact Gamma calculation identifies a compelling candidate value for the open realization step; it does not close that step. Conditional on global RH, Section 8.4 identifies the finite eigenvalue-to-root transfer as a first-order divided-difference identity. That result settles the form of the root-transfer mechanism but not the fixed-cutoff convergence rate, which still requires the Galerkin estimates and limiting norming control stated below.

Lemma 1 (Lattice sampling of the coefficients)**.** *The Fourier–Mellin transform of the N-mode eigenvector has the closed form $\widehat{\xi}_N(z) = 2L^{-1/2}\sin(zL/2)\sum_{|j|\le N}\xi_j/(z-2\pi j/L)$ ([1, Prop. 5.9]; rederived from scratch in Appendix A). On the frequency lattice $z_k = 2\pi k/L$,*

$$\widehat{\xi}_N(2\pi k/L) = (-1)^k\sqrt{L}\,\xi_k \quad (|k|\le N), \qquad \widehat{\xi}_N(2\pi k/L) = 0 \quad (|k|>N),$$

so $|\xi_k| = L^{-1/2}\big|\widehat{\xi}_N(2\pi k/L)\big|$, with $L = \ln\lambda^2$ the period of the basis $\{V_n\}$.

Proof. At $z_k = 2\pi k/L$ the factor $\sin(zL/2) = (-1)^k \sin\big((z-z_k)L/2\big)$ has a simple zero, so every $j \neq k$ term vanishes and the $j = k$ pole is removable: $\lim_{z\to z_k} 2L^{-1/2}(-1)^k \sin\big((z-z_k)L/2\big)/(z-z_k) = (-1)^k\sqrt{L}$. For $|k| > N$ there is no $j = k$ term and the common factor $\sin(z_k L/2) = 0$. □

Proposition 3 (Nyquist-cell parity)**.** *Assume that $\widehat{\xi}_N$ is real on the real axis and that neither lattice endpoint $\alpha_j = 2\pi j/L$ nor α_{j+1} is a zero. Let M_j be the total multiplicity of the real zeros of $\widehat{\xi}_N$ in the open cell (α_j, α_{j+1}). Then*

$$\operatorname{sgn}\big((-1)^j\xi_j\big)\big((-1)^{j+1}\xi_{j+1}\big) = (-1)^{M_j}.$$

Equivalently, the sampled sequence $(-1)^j\xi_j$ changes sign across a lattice cell exactly when that cell contains an odd total zero multiplicity.

Proof. By Lemma 1, the signs of $(-1)^j\xi_j$ and $(-1)^{j+1}\xi_{j+1}$ are the signs of the two endpoint values of $\widehat{\xi}_N$. A real zero of odd multiplicity reverses the sign of a real analytic function, while a zero of even multiplicity does not. Multiplying the sign changes across the cell gives the stated parity identity. □

This is a parity census, not an exact zero count: one and three zeros have the same parity, and an even-multiplicity zero is invisible to endpoint signs. In the critical finite CCM case, reality of the transform zeros is supplied by [1, Thm. 5.10] under its stated even-simple ground-state hypothesis.

Lemma 1 rigorously fixes the lattice spacing $2\pi/L$. If the coefficient tail inherits the candidate Gamma envelope, $\exp(-\delta(2\pi/L)N)$, and if the selected-zero displacement has the observed doubled exponent, then the leading digit slope is

$$\frac{\partial D_{\mathrm{nModes}}}{\partial N} = \frac{2\delta\,(2\pi/L)}{\ln 10} = \frac{\pi^2}{\ln 10\ \ln\lambda^2} \approx \frac{4.286}{\ln\lambda^2}, \tag{10}$$

a parameter-free conditional prediction. At $\lambda^2 = 7$, where few primes enter and $q \approx 1$, (10) gives 2.203 against a measured small-N slope of 2.264, agreement to 3% with no free parameters. At larger λ^2 the measured early slope runs above (10) exactly as the $q < 1$ concavity (below) requires.

What *is* rigorous is the final transfer from a locally uniform transform estimate to displacement of a simple zero. We record it next. The remaining analytic inputs are the coefficient law, the continuum spectral gap, and the observed doubling; the finite-section estimates needed between them are proved below.

Lemma 2 (Simple-zero displacement under uniform convergence)**.** *Let $a \in \mathbb{C}$, and let F be holomorphic on a neighborhood of the closed disc $\overline{D}(a,r)$, with a simple zero at a and no other zero in the disc. Shrink r if necessary so that $|F(z)| \geq \frac{1}{2}|F'(a)|\,|z-a|$ throughout the disc; this follows by applying continuity to the nonvanishing holomorphic quotient $F(z)/(z-a)$, extended at a by $F'(a)$. Let G be holomorphic on the disc and set $\eta := \sup_{\overline{D}(a,r)} |G-F|$. If $\rho := 2\eta/|F'(a)| < r$, then G has exactly one zero b in $D(a,r)$, and*

$$|b-a| \ \leq\ \frac{2\,\eta}{|F'(a)|}.$$

Proof. If $\eta = 0$, the conclusion is immediate. Otherwise, on $|z-a| = r$ the hypothesis $\rho < r$ gives $|F(z)| > \eta \geq |G(z)-F(z)|$, so Rouché's theorem gives G exactly one zero b in $D(a,r)$. For every s with $\rho < s < r$, the same strict inequality holds on $|z-a| = s$. Thus $b \in D(a,s)$ for every such s, and letting $s \downarrow \rho$ gives $|b-a| \leq \rho$. □

At fixed cutoff, assume $\widehat{\xi}_\infty$ has a simple isolated zero $\nu_\infty(\lambda)$ as in (H^*_λ). Applied with $a = \nu_\infty(\lambda)$, $F = \widehat{\xi}_\infty$, and $G = \widehat{\xi}_N$, the lemma converts uniform convergence near the continuum zero into finite-section zero displacement with the explicit constant $2/|\widehat{\xi}'_\infty(\nu_\infty)|$. It does not, by itself, compare $\nu_\infty(\lambda)$ with the zeta ordinate γ_1.

A second, high-value transfer runs from the coefficient tail to the transform. The finite-section eigenvector is not the truncation of the limit eigenvector, so the transform difference carries both a discarded-tail term and a retained Galerkin error. The three estimates needed for that passage follow from the fixed-cutoff matrix structure and the min–max principle; they need not be carried as an additional hypothesis.

Lemma 3 (Automatic fixed-cutoff Galerkin estimates)**.** *Fix λ^2 and write $L = \ln \lambda^2$. Let P_N project onto* $\operatorname{span}\{V_j : |j| \le N\}$ *and $Q_N = I - P_N$.*

(i) For every $r \ge 0$ and every $e \in \operatorname{ran} P_N$,

$$\sup_{|z-x|\le r} |\widehat{T}[e](z)| \le C_{L,r}\, \|e\|_2, \qquad C_{L,r} = \begin{cases} \big(\sinh(rL)/r\big)^{1/2}, & r > 0, \\ \sqrt{L}, & r = 0, \end{cases}$$

for every real center x. In particular, the transform-evaluation exponent is $s = 0$.

(ii) There is a constant $B_\lambda < \infty$, independent of N, such that the off-diagonal CCM matrix entries satisfy

$$|(QW_\lambda)_{jk}| \le \frac{B_\lambda}{|j-k|}, \qquad j \ne k.$$

Consequently,

$$\|P_N Q W_\lambda Q_N\|_{\ell^2 \to \ell^2} \le 2\pi B_\lambda.$$

If $|\xi_j^{(\infty)}| \le C e^{-c|j|^q}$, then for some fixed $m = \max\{(1-q)/2, 0\}$,

$$\|P_N Q W_\lambda Q_N \xi^{(\infty)}\|_2 \le C_1 N^m e^{-cN^q}.$$

(iii) Suppose the continuum ground eigenvalue ε^∞ is simple and the next spectral value is $\varepsilon_2^\infty = \varepsilon^\infty + g$ with $g > 0$. If ε_N is the lowest Ritz value of $QW_\lambda^N = P_N Q W_\lambda P_N$, then, for all sufficiently large N, it is simple and

$$\operatorname{dist}\big(\varepsilon^\infty, \sigma(QW_\lambda^N) \setminus \{\varepsilon_N\}\big) \ge g.$$

Proof. For (i), write the Fourier–Mellin transform on the logarithmic interval: $\widehat{T}[e](z) = \int_{-L/2}^{L/2} e(t) e^{izt}\, dt$. If $y = \operatorname{Im} z$, Cauchy–Schwarz gives

$$|\widehat{T}[e](z)|^2 \le \|e\|_2^2 \int_{-L/2}^{L/2} e^{-2yt}\, dt = \|e\|_2^2 \frac{\sinh(|y|L)}{|y|},$$

with limiting value $L\|e\|_2^2$ at $y = 0$. Since $|y| \le r$ on a disc centered on the real axis, the stated bound follows.

For (ii), the explicit off-diagonal formulas of [1, Secs. 4.1–4.3] write the pole, archimedean, and each of the finitely many prime-power contributions as divided differences $(b_j - b_k)/(j-k)$. At fixed λ, the corresponding source sequences are uniformly bounded. This is immediate from the rational pole profile and the finite trigonometric prime profiles. For the archimedean source, [1, eq. (4.12)] gives

$$\alpha_L(n) = \frac{1}{\pi} \int_0^L \sin(2\pi n x/L)\, \rho(x)\, dx, \qquad \rho(x) = \frac{e^{x/2}}{e^x - e^{-x}}.$$

Write $\rho(x) = 1/(2x) + h_L(x)$, where $h_L \in L^1(0, L)$. The singular part is $\frac{1}{2}\,\mathrm{Si}(2\pi n)$ and is uniformly bounded in n, while the h_L part is bounded by $\|h_L\|_1$. Thus $\sup_n |\alpha_L(n)| < \infty$. The cutoff-free formulas of [8] give the same bounded source profile. Summing the finitely many contributions gives the constant B_λ. Split the cross block into $k > N$ and $k < -N$. After reindexing, each absolute kernel is a rectangular truncation of the Hilbert matrix $(a+b+1)^{-1}$, whose ℓ^2 norm is at most π. Entrywise domination therefore gives the operator bound $2\pi B_\lambda$. Finally,

$$\|Q_N \xi^{(\infty)}\|_2^2 \le 2C^2 \sum_{j>N} e^{-2cj^q} \le C' N^{\max\{1-q,0\}} e^{-2cN^q},$$

and the residual estimate follows.

For (iii), the min–max principle gives $\varepsilon_{2,N} \ge \varepsilon_2^\infty = \varepsilon^\infty + g$ for the second Ritz value. The lowest Ritz values decrease to ε^∞ by [1, Prop. 3.4]; hence for all sufficiently large N, $\varepsilon_N < \varepsilon^\infty + g$. The first Ritz value is then simple and every other Ritz value lies at distance at least g from ε^∞. □

Lemma 4 (Tail-to-transform transfer from coefficient decay and a continuum gap)**.** *Fix λ^2. Assume that the continuum ground state of QW_λ is simple with spectral gap $g(\lambda^2) > 0$, that $\widehat{\xi}_\infty$ has a simple isolated zero $\nu_\infty(\lambda)$, and that the normalized continuum eigenvector satisfies $|\xi_j^{(\infty)}| \le C\exp(-c|j|^q)$ for some $c > 0$ and $q > 0$. Let $r > 0$ be small enough that $\overline{D}(\nu_\infty, r)$ contains no other zero of $\widehat{\xi}_\infty$ and has positive distance from the frequency lattice $\{2\pi j/L\}$. Normalize the finite ground states and fix phases by $\langle \xi^{(N)}, \xi^{(\infty)} \rangle \ge 0$. Then*

$$\sup_{|z-\nu_\infty| \le r} |\widehat{\xi}_N(z) - \widehat{\xi}_\infty(z)| \le \exp\bigl(-cN^q + o(N^q)\bigr), \qquad N \to \infty.$$

Proof. The coefficient bound is summable, so the closed form of Lemma 1 extends to the limit and defines an entire function by local uniform convergence. Split

$$\widehat{\xi}_N - \widehat{\xi}_\infty = \widehat{T}[\xi^{(N)} - P_N \xi^{(\infty)}] - \widehat{T}[Q_N \xi^{(\infty)}].$$

For the tail piece, the positive distance of the disc from the lattice and the same direct series estimate as above give

$$\sup_{|z-\nu_\infty| \le r} |\widehat{T}[Q_N \xi^{(\infty)}](z)| \le C_3 N^{-q} e^{-cN^q}(1 + o(1)).$$

For the retained piece, the Galerkin residual identity is

$$(QW_\lambda^N - \varepsilon^\infty) P_N \xi^{(\infty)} = -P_N Q W_\lambda Q_N \xi^{(\infty)}.$$

Let Π_N be the spectral projection onto the finite ground state. By Lemma 3(ii)–(iii),

$$\|(I - \Pi_N) P_N \xi^{(\infty)}\|_2 \le \frac{1}{g(\lambda^2)} \|P_N Q W_\lambda Q_N \xi^{(\infty)}\|_2 \le \mathrm{poly}(N) e^{-cN^q}.$$

After phase alignment and normalization, the same bound holds for $\|\xi^{(N)} - P_N \xi^{(\infty)}\|_2$. Lemma 3(i) converts this ℓ^2 estimate into a uniform transform estimate with no additional power of N. Polynomial factors contribute only $o(N^q)$ in the exponent. □

Corollary 1. *Under the hypotheses of Lemma 4,*

$$|\nu_1(N, \lambda^2) - \nu_1(\infty, \lambda^2)| \le \exp\bigl(-cN^q(1 + o(1))\bigr).$$

Thus a sharp coefficient tail and a continuum gap force selected-zero convergence at the same exponent, by Lemmas 4 and 2.

The transfer mechanism now has no independent Galerkin assumption. Its open inputs are the sharp continuum coefficient tail and, outside the regimes where it is known or certified, the simple isolated continuum ground state and gap. The following conjecture identifies the tail, its archimedean rate, and the observed doubling beyond Corollary 1.

Conjecture 1 (Coefficient decay and zero displacement)**.** *Fix λ^2, let $q = q(\lambda^2)$ be the exponent of Section 5.3, and let $\xi^{(\infty)}$ denote the limit eigenvector at cutoff λ^2.*

(i) (Sharp coefficient tail, full form.) *There exist $c(\lambda^2) > 0$ and C with*

$$|\xi_k^{(\infty)}| \le C \exp\bigl(-c(\lambda^2)k^q\bigr).$$

(ii) (Leading rate.) *$c(\lambda^2)L/(2\pi) \to \delta = \pi/4$ as $\lambda^2 \to \infty$, with δ the $\Gamma_{\mathbb{R}}$ envelope rate of Section 5.2. The approach is from below.*

(iii) (Doubling at the zero.) *The first-zero displacement obeys*

$$|\nu_1(N, \lambda^2) - \nu_1(\infty, \lambda^2)| \le C \exp\bigl(-2c(\lambda^2)N^q(1 + o(1))\bigr),$$

twice the tail exponent of (i).

Clause (i) is an ordinary Gevrey statement only when $q \le 1$. Clauses (ii) and (iii) together recover the parameter-free slope (10) at $q \approx 1$, validated to 3% at $\lambda^2 = 7$. The measured finite-cutoff coefficient rates remain evidence for the proposed approach toward $\pi/4$, not a proof of it. Conditional on the continuum gap, Lemma 4 and Corollary 1 show that clause (i) implies zero displacement at the single exponent $c(\lambda^2)N^q$. Clause (iii) contains the additional factor two. Conditional on global RH, Proposition 9 identifies the finite eigenvalue-to-root transfer as first order in ε_N with two powers of a divided-difference norming factor. The strict fixed-cutoff doubling claim still requires the subtracted $N \to \infty$ identity and control of $\varepsilon_N - \varepsilon^\infty$ and the limiting norming terms. No classical result supplies the $\Gamma_{\mathbb{R}}$ envelope for the finite CCM ground state; the coefficient class and its rate remain new analysis. *Status: T3. The transfer from a proved coefficient tail to a selected simple zero is T1 conditional on the continuum gap; Conjecture 1 contains the remaining coefficient-rate and doubling claims.*

5.3 The prime correction: $q(\lambda^2)$

The archimedean candidate gives the baseline $q = 1$. The arithmetic part of the localized Weil form is exactly a finite prime-power translation/cosine operator, with frequencies $k \log p$ and weights involving $(\log p)p^{-k/2}$ for $p^k \le \lambda^2$. It is *not* known that the CCM ground-state transform factorizes as a Gamma factor times a truncated Euler product. To formulate a testable fluctuation model without assuming such a factorization, isolate the first prime harmonics $S_1(z) = \sum_{p \le \lambda^2} p^{-1/2} \cos(z \ln p)$. Their long-interval mean square is

$$\langle S_1^2 \rangle = \tfrac{1}{2} \sum_{p \le \lambda^2} p^{-1} = \tfrac{1}{2}(\ln\ln \lambda^2 + M) + o(1)$$

by Mertens' theorem (M the prime Mertens constant). A separate classical identity, Mertens' product theorem, gives $\sum_{p \le \lambda^2} -\ln(1 - 1/p) = \gamma + \ln\ln \lambda^2 + o(1)$. The data show the fitted exponent decreasing as prime content grows. Motivated by these two scales, but not derived from them, model the reciprocal exponent as a baseline plus the prime fluctuation,

$$\frac{1}{q} = c_0 + c_1 \ln\ln \lambda^2, \qquad c_0 = \gamma \text{ (candidate baseline)}, \quad c_1 = \tfrac{1}{2} \text{ (candidate variance coefficient)}, \tag{11}$$

Table 1: Per-λ^2 fits of (6) (full tanh form, mode-limited points). The onset constant c clusters near $4/\pi = 1.273$; the exponent q decreases through 1 near $\lambda^2 \approx 13$, matching the archimedean crossover. The 150 and 200 columns are retained as historical reference values, but their exact ramp configurations are not preserved; both are excluded from the quantitative regression.

λ^2	7	13	20	25	50	100	150†	200†
q	1.185	0.985	0.900	0.880	0.835	0.785	0.760	0.805
c	1.20	1.25	1.28	1.29	1.31	1.28	1.26	1.31

†Historical reference value; exact ramp provenance is pending and the value is not used in regression or formal validation. Its numerical consistency with the later large-cutoff effective range does not restore that missing provenance.

which gives $q(\lambda^2) = 2/(2\gamma + \ln\ln\lambda^2)$. In this model the numerator 2 is $1/c_1$, while Mertens supplies the $\ln\ln$ scale. Neither Mertens theorem supplies the missing operator map from prime fluctuations to a sharp coefficient-decay index. The fixed operational expression is used on the calibrated domain $7 \le \lambda^2 \le 100$. Its denominator need not even remain positive as $\lambda^2 \downarrow 1$, and the large-cutoff test of Section 5.5 shows that it also cannot be extrapolated unchanged as $\lambda^2 \to \infty$.

We test (11) against per-λ^2 fits of the full form (6) (Table 1). Restricting the quantitative regression to the six cutoffs whose complete ramp configurations are preserved by the reproduction suite, a linear regression $1/q = c_0 + c_1 \ln\ln\lambda^2$ over $\lambda^2 \in \{7, 13, 20, 25, 50, 100\}$ yields

$$c_0 = 0.545 \ (\gamma = 0.577), \qquad c_1 = 0.489 \ (\tfrac{1}{2} = 0.500), \qquad R^2 = 0.977.$$

Euler–Mascheroni γ appears as the intercept of a regression that does not assume it. The offset of the parameter-free curve is systematic, not centered scatter. Taking $c_0 = \gamma$ exactly, the idealized $q = 2/(2\gamma + \ln\ln\lambda^2)$ runs 5–7% *below* the per-λ^2 fits across the whole range (e.g. 1.10 vs 1.185 at $\lambda^2 = 7$; 0.75 vs 0.785 at $\lambda^2 = 100$), and the regression intercept 0.545 sits about 5.6% under $\gamma = 0.577$, always on the same side. We therefore report both curves side by side — the provenance-complete fitted $(c_0, c_1) = (0.545, 0.489)$ and the idealized $(\gamma, \frac{1}{2})$ — and the identification of the intercept with γ is a structural hypothesis the data supports to that one-signed accuracy, not a measurement centered on it. *Status: T3; the remaining lemma is a rigorous variance-to-exponent map, now precisely posed on a Fourier coefficient-decay scale (Section 5.1): prove that the almost-periodic prime multiplier S_1, of mean square $\frac{1}{2}\sum_{p\le\lambda^2} p^{-1}$, changes the sharp exponent of the strip-analytic archimedean part according to $1/q = \gamma + \frac{1}{2}\ln\ln\lambda^2$. Where $q \le 1$, the characterization of $G^{1/q}$ by coefficient decay $\exp(-c|k|^q)$ [30] gives the standard Gevrey formulation; where $q > 1$, the corresponding ultra-analytic sequence estimate must be stated separately. What is missing in either regime is the subordination mechanism, a piece of new analysis rather than a citation.*

5.4 The Shannon coordinate

Lemma 5 (Conditional linear crossover scale)**.** *Assume the candidate leading prime ceiling and the candidate small-N linear slope of Sections 6 and 5.2. Their linear intersection occurs at*

$$N_{\text{lin}} = \frac{D_{\text{Primes}}}{\partial D/\partial N} = \frac{(4\pi/\ln 10)\lambda^2}{\pi^2/(\ln 10 \, \ln\lambda^2)} = \frac{4}{\pi}\,\lambda^2 \ln\lambda^2.$$

The λ^2 comes from the prime ceiling (Section 6); the $\ln\lambda^2$ from the inverse of the archimedean slope (10). The combination $\lambda^2 \ln\lambda^2$ is therefore the natural coordinate within this two-input model. It is Shannon-like: the factor $L = \ln\lambda^2$ converts a physical-frequency scale to a number of lattice modes through the spacing $2\pi/L$. A literal identification with the

Shannon number of a specific prolate operator requires the still-open CCM/prolate comparison; the classical time–band limiting results [34, 33, 32] motivate that interpretation but do not supply it automatically.

The crossover multiplier and the fitted tanh coefficient must not be conflated. Equation (6) places c *inside* the argument, so its unit-argument scale is $N_{\tanh} = \lambda^2 \ln \lambda^2 / c$. Thus the observation $c \approx 4/\pi$ does not follow from $N_{\mathrm{lin}} = (4/\pi)\lambda^2 \ln \lambda^2$; one occurrence is a fitted argument coefficient and the other is a crossover multiplier. Their relation also depends on the stretched exponent and the chosen saturation convention. Measured saturation occurs at $N/(\lambda^2 \ln \lambda^2) \approx 2.0$–$2.3$, while the fitted onset coefficient clusters near $c \approx 1.27$ (Table 1). A second, independent motivation comes from the geometric capacity derived in Section 8.2. Ignoring the 7/8 and oscillatory terms and setting $K^{(0)}_{\max}(N, \lambda^2) \approx N$ gives

$$\log\left(\frac{N}{\ln \lambda^2}\right) - 1 \approx \ln \lambda^2, \qquad \text{hence} \qquad N \approx e\, \lambda^2 \ln \lambda^2.$$

This is a constraint-counting heuristic: it says that the transition should occur when the number of geometrically available ordinate constraints is of the same order as the basis dimension. A second comparison uses local densities. The retained frequency lattice has density $L/(2\pi)$ per unit height, while the smooth zeta-zero density at $T_{\max}$ is $\frac{1}{2\pi} \ln(T_{\max}/2\pi) = \frac{1}{2\pi} \ln(N/L)$. Equating the two gives $N \approx \lambda^2 \ln \lambda^2$, whereas equating cumulative capacity to the basis dimension gives $N \approx e\lambda^2 \ln \lambda^2$. Thus the two independent geometric comparisons place the transition on the corridor

$$\lambda^2 \ln \lambda^2 \;\lesssim\; N \;\lesssim\; e\lambda^2 \ln \lambda^2.$$

This derives the coordinate and an $O(1)$ transition scale from the paper's own capacity formula, but it does not prove independence of the constraints, identify the empirical tanh convention, or derive its onset coefficient. Deriving c, or even proving its exact value is $4/\pi$, remains open. *Status: the algebra of the crossover lemma is T1 conditional on its T3 rate inputs; the capacity crossover is heuristic; the coordinate and the fitted onset constant remain T3.*

5.5 Operational validation and large-cutoff constraint

The accuracy of the operational formula must be assessed with its constants fixed. Per-cutoff refits test the flexibility of the stretched-tanh family; they do not test the operational formula itself. We therefore report the fixed submodel

$$D_1^{\mathrm{pred}}(\lambda^2, N) = D_{\mathrm{Primes}}(\lambda^2) \tanh\left[\left(\frac{(4/\pi)N}{\lambda^2 \ln \lambda^2}\right)^{q_\gamma(\lambda^2)}\right], \quad D_{\mathrm{Primes}} = \frac{4\pi}{\ln 10}\lambda^2 - 5 \log_{10} \lambda^2 - 9.5.$$

On an independent exact-assembly reimplementation, the fixed formula gives the following errors; residual means formula minus measurement.

Table 2: Cross-implementation validation of the fixed published formula on the calibrated domain. The independent implementation does not refit c, q_γ, or the cutoff ceiling at any displayed cutoff; all values use the published fixed constants.

λ^2	points	measured span	mean residual	RMS	maximum
7	23	15.14	+0.33	0.47	0.87
13	15	43.29	+0.12	0.66	1.07
20	16	74.36	−0.61	1.18	2.12

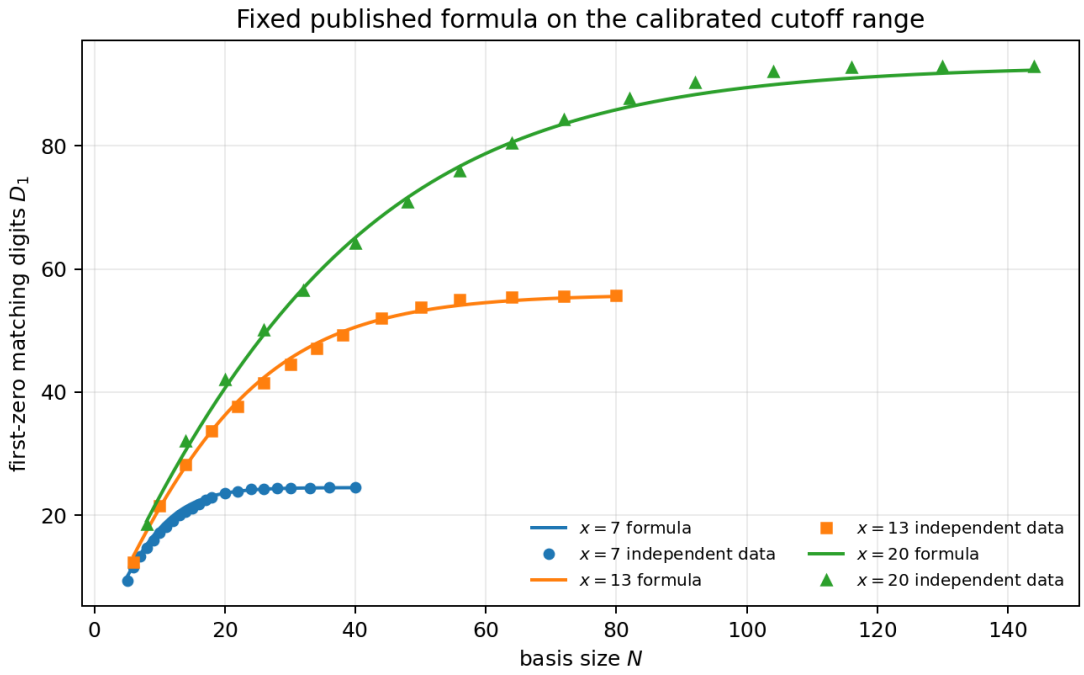


Figure 1: Independent measurements and the operational formula with all published constants fixed. The close agreement tests the formula, not merely the flexibility of the tanh family.

Paper 1 supplies a separate fixed-$N = 120$ cross-check. At $\lambda^2 = 20, 30, 50, 100$, the fixed formula has signed residuals $-1.54, -1.66, +2.13, +6.03$ digits against measured values ranging from 92.817 to 211.29 digits [51]. The growing positive tail is small relative to the total depth, but it is the first visible warning that the closed parameter map is beginning to drift.

Independently, Groskin reports eigenvalue depths at $\lambda^2 = 100$ for $N = 100, 150, 200, 250$ [7]. A global log–log fit to those public values gives $q_{\text{eff}} = 0.736$, compared with $q_\gamma(100) = 0.746$. Holding the published $c = 4/\pi$ and ceiling fixed gives residuals of approximately $+1.5, +3.7, +4.0, +3.4$ digits over a 143-digit ramp. This independently supports the calibrated exponent at $\lambda^2 = 100$. It tests the eigenvalue-depth ramp, not the first-root digits; the finite conversion factor of Section 8.4 separates the two observables.

That warning becomes decisive at $\lambda^2 = 1000$. A four-point sweep, independently reproduced by the Rust implementation, gives:

Table 3: Large-cutoff test of the unchanged closed submodel. The conversion gap is $D_{\text{eig},N} - D_1$; residual is formula minus measured D_1.

N	$D_{\text{eig},N}$	D_1	conversion gap	formula	residual
60	180.88	175.36	5.52	293.24	+117.88
90	247.60	243.06	4.54	381.08	+138.02
120	308.77	304.69	4.08	458.81	+154.12
150	366.41	362.60	3.81	529.75	+167.15

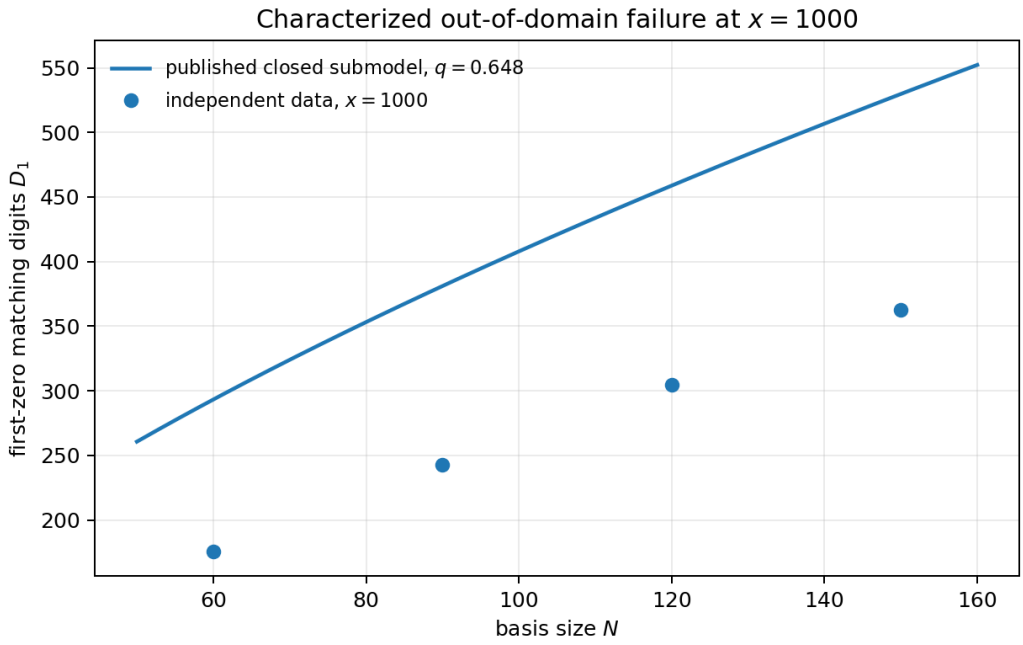


Figure 2: The closed calibrated-range parameter map overpredicts the new $\lambda^2 = 1000$ ramp. This is a characterized out-of-domain failure of q_γ and $c = 4/\pi$, not a failure of the minimum-budget structure or of the stretched-transition family.

The inferred effective exponent is convention dependent: a log–log fit of D_1 gives 0.793, a fit of $D_{\mathrm{eig},N}$ gives 0.770, fixing $c = 4/\pi$ and inverting the level at $N = 120$ gives 0.756, and jointly fitting (c, q) gives $(1.512, 0.792)$. A wide-decade eigenvalue-depth comparison to the archived $N = 800$ point gives approximately 0.743. The robust conclusion is therefore only

$$q_\gamma(1000) = 0.648 \text{ is too small}, \qquad q_{\mathrm{eff}}(1000) \approx 0.74\text{–}0.80,$$

while the data do not yet decide whether the missing correction belongs to q, to c, or to the transition coordinate. The next formula-selection pass will compare, by whole-cutoff holdout: the present q_γ model; the fitted linear law for $1/q$; a saturated exponent; a logarithmically corrected Kulikov-type plunge coordinate; and a two-stage eigenvalue-depth minus conversion-gap model. No replacement enters the boxed law until it improves whole-cutoff holdouts while preserving the calibrated-domain accuracy.

Status: the fixed-form validation and the $\lambda^2 = 1000$ failure are T3 numerical measurements. The four large-cutoff points were independently reproduced by the Rust implementation to the displayed precision.

6 The prime ceiling D_{Primes}

The prime channel of the intrinsic law (1) is the accuracy at infinite N and P, set solely by which primes $p \le \lambda^2$ enter the form. It is the relative error of the first zero at the cutoff,

$$\varepsilon_{\mathrm{primes}}(\lambda^2) = \frac{|\nu_1(\infty, \lambda^2) - \gamma_1|}{|\gamma_1|}, \tag{12}$$

with $\nu_1(\infty, \lambda^2)$ the first zero at infinite basis size and fixed λ^2: it is what remains once modes and precision are exhausted, and its digit measurement $D_{\mathrm{Primes}} = -\log_{10} \varepsilon_{\mathrm{primes}}$ is the ceiling the minimum (2) consumes. We fix its leading rate through the plunge *eigenvalue*, the tractable route, kept here in digit space to reinforce the intrinsic statement with the measured quantity.

The plunge eigenvalue gives an *eigenvalue floor* $D_{\mathrm{eig}} = -\log_{10}\varepsilon^{\infty}(\lambda)$, the digit value of the smallest eigenvalue. Its prolate comparison requires two indexing conventions to be kept separate. Let $\mu_n(c)$ denote the finite-Fourier singular value in the full PSWF indexing and let $\Lambda_n(c) = \mu_n(c)^2$ be the corresponding concentration eigenvalue. The function $h_{4,\lambda}$ has full index $n = 4$; in the even-sector notation of Connes [4, §6.3], its positive finite-Fourier eigenvalue is $\chi_2 = \mu_4$. Thus it is not the principal $n = 0$ quantity χ_0.

The classical finite-Fourier concentration eigenvalues are simple and strictly ordered; Slepian–Pollak's original argument uses endpoint nondegeneracy and a same-parity Wronskian identity [33, Sec. VI, eqs. (27) and (38)]. The exact trace identity of Landau–Pollak gives the corresponding phase-space count and is the primary source of the Shannon-number terminology [34, eq. (4.1)]. These results establish the classical prolate structure only. They do not identify the CCM ground state or transfer its convergence rate.

Fuchs' fixed-index theorem [31], at $c = 2\pi\lambda^2$, gives

$$1 - \Lambda_4(c) \sim \frac{2^{15}\sqrt{2}\,\pi^5}{3}\,(\lambda^2)^{9/2} e^{-4\pi\lambda^2}.$$

Since $\mu_4 = \sqrt{\Lambda_4} \to 1$, $1 - \mu_4 = (1 - \Lambda_4)/(1 + \mu_4)$, and therefore

$$1 - \chi_2(2\pi\lambda^2) \sim \frac{2^{14}\sqrt{2}\,\pi^5}{3}\,\exp\!\big(-4\pi\lambda^2 + \tfrac{9}{2}\ln\lambda^2\big). \tag{13}$$

Equivalently, the classical prolate digit scale is

$$\begin{aligned} D_{\mathrm{prol}}(\lambda^2) &:= -\log_{10}(1 - \chi_2(2\pi\lambda^2)) \\ &= K\lambda^2 - \frac{9}{2\ln 10}\ln\lambda^2 - C_0 + o(1), \qquad K = \frac{4\pi}{\ln 10} = 5.45750\ldots. \end{aligned} \tag{14}$$

where $C_0 = \log_{10}\!\big(2^{14}\sqrt{2}\,\pi^5/3\big) = 6.3736\ldots$. This fourth-index singular-value asymptotic is classical and T1. Groskin [7] independently evaluates the same §6.4 prefactor and obtains $\log_{10} \approx 6.37$. Hedenmalm's independent theta-kernel formulation contains the related prefactor $t^{9/2}e^{-\pi t^2}$ in its large-t asymptotic [6, Eq. (1.2.3)]; it does not supply the CCM-to-prolate transfer. The separate assertion that the CCM eigenvalue floor D_{eig} follows this scale is the prolate-transfer step discussed next; it is strongly supported by the empirical data but is not part of Fuchs' theorem.

The variational part of the proposed transfer can be anchored rigorously, and we record exactly that part as a proposition. Recall that the bottom of the spectrum of the continuum form is a variational infimum, $\varepsilon^{\infty}(\lambda) = \inf_{\varphi} QW_{\lambda}(\varphi,\varphi)/\|\varphi\|^2$, so *any* trial vector bounds it from above, irrespective of the sign of ε^{∞}.

Proposition 4 (Variational trial bound)**.** *Let h_{λ} be the suitably normalized, zero-integral linear combination of $h_{0,\lambda}$ and $h_{4,\lambda}$ used in the CCM prolate construction, and let $k_{\lambda} = E(h_{\lambda})$ be its image under the CCM Poisson-summation map ([1, §7]; [4, §6.4]). Then, unconditionally,*

$$\varepsilon^{\infty}(\lambda) \;\le\; R[k_{\lambda}] := \frac{QW_{\lambda}(k_{\lambda},k_{\lambda})}{\|k_{\lambda}\|^2}.$$

Proof. The bottom of the spectrum is the infimum of the Rayleigh quotient over the form domain; any admissible trial vector bounds the infimum from above. □

The proposition is the variational principle verbatim and is T1. It does not by itself identify the size of $R[k_{\lambda}]$. On the localized form's exact split $QW_{\lambda} = A_{\mathrm{arch}} - A_{\mathrm{prime}}$ (archimedean part $w_{02} - w_r$, prime part the prime-power sum w_p; [1, §6]), define

$$R[k_{\lambda}] = R_{\mathrm{arch}}[k_{\lambda}] - \rho_p[k_{\lambda}], \qquad \rho_p[k_{\lambda}] := \frac{\langle A_{\mathrm{prime}}k_{\lambda}, k_{\lambda}\rangle}{\|k_{\lambda}\|^2}. \tag{15}$$

Lemma 6 (Conditional CCM-to-prolate floor bound)**.** *Suppose, as $\lambda \to \infty$, that the constrained CCM trial vector satisfies*

$$R_{\mathrm{arch}}[k_\lambda] \le (1 - \chi_2(2\pi\lambda^2))(1 + o(1)) \quad \textit{and} \quad \rho_p[k_\lambda] \ge 0.$$

Then

$$\varepsilon^{\infty}(\lambda) \le (1 - \chi_2(2\pi\lambda^2))(1 + o(1)).$$

Proof. By Prop. 4 and (15), $\varepsilon^{\infty}(\lambda) \le R[k_\lambda] \le R_{\mathrm{arch}}[k_\lambda]$ by the prime-sign hypothesis, and the stated comparison with $1 - \chi_2$ finishes the proof. □

The first hypothesis is the currently open CCM-to-prolate quotient comparison. Connes [4, §6.4] reports the numerical similarity between $\varepsilon^{\infty}(\lambda)$ and $1 - \chi_2(\lambda)$ and describes k_λ as an educated approximation to the ground state; neither statement is an exact Rayleigh-quotient identity. The second hypothesis is a sign on the prime quotient evaluated on this same constrained trial vector. The paper's measured prime contributions on the computed ground state support the sign but do not prove it for k_λ. Andrade's exact Loewner formula proves a termwise non-positive prime contribution for the specific even vector $\cosh(x/2)$ and exposes mixed signs in the odd sector [10]. It does not prove the sign for the constrained h_0–h_4 vector k_λ, but it sharpens the open task: control the sign or weighted correlation of the exact Fourier symbol of this particular trial state, rather than estimate the prime matrix entrywise.

Under both hypotheses, Eq. (13) gives the conditional one-sided floor bound

$$D_{\mathrm{eig}}(\lambda^2) \;\ge\; K\lambda^2 - \frac{9}{2\ln 10}\ln\lambda^2 - C_0 + o(1), \qquad K = \frac{4\pi}{\ln 10},$$

whenever $\varepsilon^{\infty}(\lambda) > 0$. The empirical CCM calculations used throughout this paper are unchanged and align with this fourth-index law; what remains open is its analytic transfer from the constrained h_0–h_4 trial construction to the full CCM floor. The reverse inequality, that the ceiling cannot be beaten, i.e. a *lower* bound on ε^{∞} and on $|\nu_1(\infty, \lambda^2) - \gamma_1|$ of the same exponential order, remains open, and is the archimedean–prime cancellation of Section 9.

Where finite-λ^2 control (rather than asymptotics) is needed, for instance the curvature that produces the empirically steeper matching-floor slope $P_m \approx 5$ below, the modern non-asymptotic prolate literature supplies two-sided eigenvalue bounds with explicit constants: the three-region estimates of Bonami–Jaming–Karoui [22], the improved bounds of Karnik–Romberg–Davenport [23], the upper bounds of Osipov [24], the super-exponential decay estimates of Bonami–Karoui [25], and Kulikov's exponential lower bound before the plunge region [26]. These replace the asymptotic-only inputs wherever a quantified inequality at a specific cutoff is wanted.

The eigenvalue depth is a scaffold, not the object. What the law consumes is the zero error $\varepsilon_{\mathrm{primes}}$, which sits a few digits below the eigenvalue floor: converting the plunge depth ε^{∞} into the zero displacement $|\nu_1 - \gamma_1|$, and normalizing by $|\gamma_1|$ (the $\log_{10}|\gamma_1| \approx 1.15$ digits of the relative error), gives the matching floor

$$D_{\mathrm{Primes}}(\lambda^2) = K\lambda^2 - P_m \log_{10}\lambda^2 - c_m, \qquad P_m \approx 5, \quad c_m \approx 9.5. \tag{16}$$

The leading $K\lambda^2$ is shared with the eigenvalue floor, the same plunge rate, so the intrinsic content is a *single* channel, not two competing ceilings; the subleading P_m, c_m are the eigenvalue-to-zero conversion together with the $|\gamma_1|$ normalization. The expanded artifact archive permits a more direct test of that conversion. For each numerical configuration define

$$\mathrm{conv}_N(\lambda^2) := -\log_{10}|\varepsilon_N| - D_N,$$

pairing the stored smallest Weil eigenvalue with the stored relative error of the first refined root. On the deepest precision-clear configuration at each of thirteen cutoffs, this gap grows

from 2.499 digits at $\lambda^2 = 7$ to 2.925 at 100, then is 3.024–3.029 throughout $\lambda^2 = 500$–1200. Thus the earlier algebraic extrapolation $(P_m - \frac{9}{2}) \log_{10} \lambda^2 + (c_m - C_0)$ is not supported as a growing conversion law. The paired data instead favor a conversion approaching a constant $C_{\mathrm{conv}} \approx 3.03$ digits. If this persists in the saturated continuum limit, then the asymptotic logarithmic coefficient returns to $P_m = 9/2$, while $c_m = C_0 + C_{\mathrm{conv}} \approx 6.3736 + 3.03 = 9.40$, close to the finite-range fit $c_m \approx 9.5$. This is a sharper hypothesis, not yet a theorem: several large-cutoff configurations are mode-limited, and a uniform target-transform comparison is still required. The leading rate K is derived for the classical fourth-index deficiency and is observed for the CCM floor; the transfer between those two quantities is the conditional step in Lemma 6. The finite-range fit, hence P_m and c_m, remains empirical; of it, the ≈ 1.15-digit part of c_m is the explicit $|\gamma_1|$ normalization and the remainder is the depth-to-zero factor. Conditional on global RH, the finite conversion factor is no longer merely a fitted constant: equation (32) below expresses $D_{\mathrm{eig},N} - D_k$ exactly through the finite Stieltjes factor, divided difference, and remote-zero term. That conditional finite identity does not prove the saturated continuum limit, the large-cutoff constancy of C_{conv}, or the unconditional comparison needed in an RH argument. A structural unconditional route to the limiting factor would require an additional target-transform comparison, which we record as a proposal rather than claim as a derivation. Specifically, let Z_λ be a compatibly normalized holomorphic target transform on a disc about γ_1, with a simple zero at γ_1, and suppose one can prove

$$\eta_\lambda := \sup_{\overline{D}(\gamma_1, r)} |\widehat{\xi}_\infty(z) - Z_\lambda(z)| \quad \text{is sufficiently small.}$$

Lemma 2, now applied with $a = \gamma_1$, $F = Z_\lambda$, and $G = \widehat{\xi}_\infty$, would then give

$$|\nu_\infty(\lambda) - \gamma_1| \le \frac{2\eta_\lambda}{|Z'_\lambda(\gamma_1)|}.$$

The determinant identity $\det_{\mathrm{reg}}(D - z) = -i\lambda^{-iz}\widehat{\xi}(z)$ of [1, Thm. 5.10] describes the finite transform, but does not itself construct Z_λ or prove this uniform estimate. Establishing that comparison, connecting η_λ to the plunge depth, and deriving the λ-scaling of the ratio would promote (P_m, c_m) from empirical to structural. Until then the row remains T4.

The data indicate that the arithmetic does not move the eigenvalue-floor slope. A formal large-$|t|$ model motivated by Mertens' theorems and prime-number-theorem partial summation has the scalar form $S(t) = \frac{1}{2}t^2 - \gamma|t| + c_0$. Turning this scalar asymptotic into a statement about the compressed CCM prime operator requires its own operator limit: parity, the mean-zero constraint, and the compression Q do not by themselves justify assigning the three displayed terms separate actions on the finite-section ground state. The screw-function decomposition of Suzuki [5] suggests a useful comparison with the metric operator, while the exact finite prime-power translations provide a second, independent route through quadratic-form or Loewner estimates. Either route could explain why the arithmetic changes lower-order conversion terms without changing the prolate prefactor power, but that conclusion is not yet a theorem. Conditional on the transfer in Lemma 6, the prefactor power is the Fuchs index $\frac{9}{2}$ of the fourth-index prolate model (Section 6), while the arithmetic contribution enters the lower-order conversion. The archimedean part of the form does not by itself carry the plunge (it is $O(1)$-indefinite; see the rate-origin discussion in Section 9). The empirically steeper matching-floor slope $P_m \approx 5$ is a finite-λ^2 curvature effect, not a change in the asymptotic slope.

Status: the classical fourth-index prolate coefficient K and slope $\frac{9}{2}$ are T1. Their transfer to the CCM eigenvalue floor is T3: it is validated by the data, while the constrained-trial comparison and the prime sign in Lem. 6 remain open. The reverse inequality requires new archimedean–prime cancellation estimates. The matching-floor offset (P_m, c_m), which also contains the eigenvalue-to-zero conversion, remains empirical (T4). The paired artifacts now isolate a candidate constant $C_{\mathrm{conv}} \approx 3.03$ and favor the asymptotic value $P_m = 9/2$; the structural route remains the target-transform comparison named above.

A smooth envelope with arithmetic microstructure, not a staircase. A natural hypothesis, that D_{Primes} changes by macroscopic steps when a prime power enters, is falsified directly. With N and P held above floor ($N = 300$, $P = 200$), sweeping λ^2 in unit steps from 4 to 30 gives an overall rise of ≈ 5.3 digits per unit λ^2; at that resolution the per-unit rate at prime-power entries is statistically indistinguishable from non-prime entries. This does not mean the exact matrix path is smooth across an entry event. Groskin proves that its first derivative has a rank-one jump at every prime power [9]; small local bumps or kinks are therefore expected. The empirical conclusion is the more precise one: Eq. (16) describes a smooth macroscopic envelope, not a distinct digit staircase, while the arithmetic events survive as finer path structure.

Proposition 5 (Prime-event slope law)**.** *Let $u = \ln \lambda^2$, let $q = p^m$ be a prime power, and set $u_q = \ln q$. Write $Q_N(u)$ for the finite CCM matrix path and assume that its selected eigenvalue $\varepsilon_N(u_q)$ is simple, with normalized real eigenvector ξ_q. Then the one-sided eigenvalue derivatives satisfy*

$$\boxed{\varepsilon'_N(u_q+) - \varepsilon'_N(u_q-) = -\frac{2\Lambda(q)}{\sqrt{q}\,\ln q}|\langle \mathbf{1}_N, \xi_q\rangle|^2,}$$

where $\mathbf{1}_N$ is the all-ones vector in $\mathbb{R}^{2N+1}$. If $\varepsilon_N(u_q) > 0$ and $D_{\mathrm{eig},N}(u) = -\log_{10}\varepsilon_N(u)$, then

$$\boxed{D'_{\mathrm{eig},N}(u_q+) - D'_{\mathrm{eig},N}(u_q-) = \frac{2\Lambda(q)}{\sqrt{q}\,\ln q\,\ln 10}\frac{|\langle \mathbf{1}_N, \xi_q\rangle|^2}{\varepsilon_N(u_q)} \geq 0.}$$

Proof. Groskin's prime-event theorem gives the exact matrix jump

$$Q'_N(u_q+) - Q'_N(u_q-) = -\frac{2\Lambda(q)}{\sqrt{q}\,\ln q}\,\mathbf{1}_N\mathbf{1}_N^T \qquad [9, Thm.\ 3.1].$$

The matrix path is continuous at u_q. Simplicity therefore gives a common normalized eigenvector at the event, and the one-sided Hellmann–Feynman identities yield the first formula after subtraction. The second follows from $D'_{\mathrm{eig},N} = -\varepsilon'_N/(\varepsilon_N \ln 10)$ and continuity of ε_N. □

With

$$w_0 = \left|\left\langle (2N+1)^{-1/2}\mathbf{1}_N, \xi_q\right\rangle\right|^2,$$

the event amplitude is

$$\Delta\varepsilon'_N = -\frac{2\Lambda(q)}{\sqrt{q}\,\ln q}(2N+1)w_0.$$

Thus each prime power leaves the matrix value continuous but produces an exact downward kink in the finite eigenvalue and an upward kink in its digit depth, with von Mangoldt weight modulated by the boundary spectral mass. This is a finite-N eigenvalue statement. It does not by itself give a derivative jump for the transform-root error or for the saturated continuum ceiling.

Floor anchors. Floor-isolated measurements (both N and P above floor) give $D_{\mathrm{Primes}}(100) = 526.08$ at $(N, P) = (1000, 700)$ and a saturated $D_{\mathrm{Primes}}(200) \approx 1070.4$ at $N \geq 2400$. The latter sits at the matching floor $K\cdot 200 - 5\log_{10} 200 - 9.5 = 1070.5$, confirming the two-floor structure at the largest cutoff measured. Regarding provenance, the $\lambda^2 = 200$ anchor was measured against the 2500-digit Arb reference of Definition 1, not a shorter tabulation. A matching-digit count cannot exceed the length of its reference: against a ~ 1000-digit reference the measurement would *saturate* near 999 and could never read 1070.4, so the observed value is itself incompatible with a reference-limited artifact. Such an artifact would present as a suspicious plateau at the reference length, not a reading 70 digits past it.

7 The Riemann–von Mangoldt backbone and the spectral-edge hypothesis

Sections 5 and 6 approach the mode ramp and the prime ceiling by two separate classical routes: the Landau–Widom transition and the Fuchs prolate tail. Their shared scales and the data suggest that they are two regimes of one CCM spectral edge. This section separates the rigorous smooth counting backbone from that stronger, offensive hypothesis.

7.1 The rigorous archimedean counting backbone

Let

$$\vartheta(T) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{iT}{2}\right) - \frac{T}{2} \log \pi$$

be the Riemann–Siegel phase in the standard continuous branch. The Riemann–von Mangoldt formula can be written

$$N_\zeta(T) = \frac{\vartheta(T)}{\pi} + 1 + S(T),$$

away from ordinates, with the usual limiting convention at a zero. Stirling's formula gives the unconditional smooth density

$$\rho^{(0)}_{\mathrm{RvM}}(T) := \frac{\vartheta'(T)}{\pi} = \frac{1}{2\pi} \log \frac{T}{2\pi} + O(T^{-2}). \tag{17}$$

Thus the archimedean Gamma phase supplies the smooth Riemann–von Mangoldt backbone, while $S(T)$ contains the oscillatory correction. This statement is independent of RH and does not require an equilibrium-measure interpretation. Evaluating the integrated smooth density at the exact finite-band edge $T_{\max} = 2\pi N / \ln \lambda^2$ recovers the leading terms of $K^{(0)}_{\max}$ in (23). The reach formula and its smooth capacity therefore remain rigorous independently of the conjectural edge picture.

The earlier formal identification of (17) with the minimizer of an unscaled unit-mass logarithmic energy is not valid: the density has infinite mass on the half-line, and differentiating a weighted-energy Euler–Lagrange condition produces a singular Hilbert-transform equation rather than $\rho = Q'/\pi$. A genuine equilibrium formulation must specify a scaling window, a normalized empirical measure, its external field, and which spectral objects are being counted. Weighted logarithmic-energy theory [35] and its orthogonal-polynomial/random-matrix realization [36] provide candidate tools once those CCM-specific objects are derived; their general results do not supply the missing identification by themselves.

7.2 A scaled equilibrium conjecture

For a fixed $0 < a < 1$, a candidate CCM zero measure in a high-frequency window is

$$\mu^{\mathrm{CCM}}_{T,a} := \frac{1}{K_{T,a}} \sum_{aT < \nu_k \le T} \delta_{\nu_k / T},$$

where $K_{T,a}$ is the number of selected transform zeros in the window. Its smooth Riemann–von Mangoldt target is the probability measure

$$d\mu^{(0)}_{T,a}(x) := \frac{T \rho^{(0)}_{\mathrm{RvM}}(Tx)\, \mathbf{1}_{[a,1]}(x)\, dx}{\int_{aT}^{T} \rho^{(0)}_{\mathrm{RvM}}(t)\, dt}.$$

These definitions do not assert that the CCM zeros form a logarithmic gas; they state a normalized object on which such a theorem could be formulated.

Conjecture 2 (Scaled CCM equilibrium and common edge)**.** *Along a joint regime in which the cutoff and finite-section reach tend to infinity:*

(i) the selected CCM zero measures have the same macroscopic limit as $\mu_{T,a}^{(0)}$*;*

(ii) there are scaled external fields for which these measures are asymptotically characterized by a weighted logarithmic-energy problem; and

(iii) the observed mode transition and prime tail arise as two scales of one associated CCM spectral edge.

The most direct offensive route uses the determinant identity $\det_{\mathrm{reg}}(D'' - z) = -i\lambda^{-iz}\widehat{\xi}(z)$. Numerical and regularized Fredholm determinant theory supplies templates for controlling finite determinants and their product anomalies [48, 49, 50]. A limit for the normalized subharmonic functions

$$\frac{1}{K}\log\left|\det_{\mathrm{reg}}(D'' - z)\right|$$

would determine a limiting zero measure through its distributional Laplacian and could identify the external field and logarithmic potential. This route addresses transform zeros directly and supplies the missing bridge between the finite determinant and a potential-theoretic measure.

A complementary route begins with the archimedean matrix. Its explicit phase rate $\vartheta'(t) = \frac{1}{2}\log(t/2\pi) + O(t^{-2})$ suggests a Toeplitz-, Kac–Murdock–Szegő-, or pseudodifferential-type spectral law [28, 29]. To make that route rigorous one must first identify the exact matrix class and normalization, handle the unbounded logarithmic symbol, verify form-core and spectral-distribution hypotheses, and obtain a uniform estimate for the arithmetic perturbation. Even then, a separate theorem must connect the resulting matrix eigenvalue measure to the transform-zero measure. The discreteness result of [1, Thm. 3.6] motivates this program but does not by itself establish all of these quantitative steps.

7.3 One-edge conjecture: two observed scales

In the parameter range where the relevant Weil form is positive, the computed spectrum descends from an $O(1)$ bulk into a deep plunge. Two classical facts make a common-edge interpretation compelling:

- the Landau–Widom transition width [32] supplies the natural crossover scale near the Shannon number $\lambda^2 \ln \lambda^2$, matching the observed mode ramp;
- the fourth-index Fuchs asymptotic [31] supplies the extreme prolate tail, matching the observed leading scale of the prime ceiling.

The classical plunge is commonly written as a Fermi–Dirac/logistic profile; after centering and rescaling, a logistic is algebraically a shifted tanh. This makes the empirical choice of a hyperbolic tangent less ad hoc, but Landau–Widom's index variable is not yet the CCM root-error coordinate. Modern pre-plunge estimates sharpen that warning: before the transition, the natural variable contains a logarithmically corrected distance to the Shannon edge rather than a pure normalized power [46, 47]. The measured failure of the unchanged q_γ model at $\lambda^2 = 1000$ therefore elevates a log-corrected plunge coordinate from an optional analogy to a leading empirical competitor. Identifying the CCM variable, including whether the correction should be assigned to q, to c, or to the coordinate itself, remains part of the open edge comparison rather than a consequence of the classical profile. The offensive hypothesis is that these are the transition and extreme tail of the same CCM edge, explaining the algebra $D_{\mathrm{nModes}} = D_{\mathrm{Primes}}\, g$. This is not yet a consequence of the Riemann–von Mangoldt bulk density: the Weil-matrix spectrum, the perturbed operator spectrum, the transform zeros, and the prolate concentration spectrum

must first be connected by explicit comparison theorems. The microscopic universality class (Airy, Bessel, or Landau–Widom), the determinant-to-measure bridge, and the exact deep-edge constant remain open. *Status: T3 — a structural conjecture supported by two classical asymptotic regimes and the measured convergence surface; proving it requires new potential-theoretic and spectral-comparison analysis.*

8 The complete formula

8.1 The assembled law

Collecting Sections 4–7, the law stands in the two forms of the introduction. *Intrinsically*, the relative error obeys the two-channel decomposition

$$E(N,\lambda^2) \le \varepsilon_{\text{modes}}(N,\lambda^2) + \varepsilon_{\text{primes}}(\lambda^2) \tag{18}$$

(an equality up to sign cancellation; Prop. 1); equivalently, in digit units, the edge function $D_{\text{edge}}(N,\lambda^2) = D_{\text{Primes}}(\lambda^2)\cdot\tanh\big((c\,N/(\lambda^2\ln\lambda^2))^q\big)$, whose $N\to\infty$ ceiling $D_{\text{edge}}(\infty,\lambda^2) = D_{\text{Primes}}$ is the prime floor and whose finite-N transition is the mode ramp D_{nModes}. *Observed* at the requested target $P = P_{\text{req}}$, with internal working precision P_{work}, the parametrized model is capped by the instrument resolution:

$$D(\lambda^2,N,P) = \min\Big(\underbrace{P_{\text{req}}}_{D_{\text{Wprec}}},\ \underbrace{D_{\text{Primes}}\tanh\big((c\,N/(\lambda^2\ln\lambda^2))^{q_\gamma}\big)}_{D_{\text{nModes}}=D_{\text{edge}}},\ \underbrace{K\lambda^2 - P_m\log_{10}\lambda^2 - c_m}_{D_{\text{Primes}}=D_{\text{edge}}(\infty)}\Big) + O(1), \tag{19}$$

with the classical candidate $K = 4\pi/\ln 10$ and, on $7\le\lambda^2\le 100$, the T3 operational submodel $q = q_\gamma = 2/(2\gamma+\ln\ln\lambda^2)$ and $c = 4/\pi$. The value of K is exact for the fourth-index prolate deficiency; its transfer to the CCM root ceiling is not. Outside the calibrated mode domain, equation (19) retains the minimum structure and stretched-transition family but does not assign a settled large-cutoff parameter map. The last two entries admit the common-edge parametrization D_{edge} at finite N and at $N\to\infty$, so the operative model is $D = \min(D_{\text{Wprec}}, D_{\text{edge}}) + O(1)$. Its interpretation as a single spectral edge is the conjecture of Section 7; the formula and measured surface do not depend on that conjecture being proved. This is the Andrews CCM Convergence Law. The law is the budget skeleton and its empirically measured channel functions; the particular closed $(q_\gamma, 4/\pi)$ map is its calibrated operational realization, not a construction-wide theorem. Of the law's parts, the minimum bound, the precision slope, the stretched-exponential rate unification (Prop. 2), and the general simple-zero displacement (Lemma 2) are rigorous; every remaining component carries the explicit tier, open item, and open-item type recorded in Table 4, from the conditional transfer estimates through the T3 exponent and onset constant to the empirical matching-floor terms (T4).

Operational model card. The following compact statement is the operational version that can be applied and independently tested.

Item	Operational statement
Inputs	cutoff λ^2, basis size N, requested decimal target P_{req}
Output	predicted relative first-zero matching digits D_1
Precision map	$D_{\text{Wprec}} = P_{\text{req}} + O(1)$; the toolkit adds a 64-bit internal reserve
Closed mode submodel	equation (6), validated for $7\le\lambda^2\le 100$
Cutoff ceiling	equation (16), calibrated and floor-tested through $\lambda^2 = 200$
Independent validation	fixed-form RMS 0.47, 0.66, 1.18 digits at $\lambda^2 = 7, 13, 20$ (Table 2)
Known failure	unchanged closed mode map overpredicts the $\lambda^2 = 1000$ ramp by 118–167 digits (Table 3)
Data admissibility	validated matrix assembly, reference provenance, structural checks, and precision/cutoff stability

To the best of the author's knowledge, the proposed complete convergence model of (19) is new: the assembly of the three budgets into a single law, the calibrated closed-form mode exponent $q_\gamma(\lambda^2) = 2/(2\gamma + \ln\ln\lambda^2)$, the Shannon-like coordinate $\lambda^2 \ln\lambda^2$, the stretched-exponential parametrization of the mode budget, and the proposed common-edge interpretation of the mode ramp and prime ceiling appear here for the first time. The rate it models is precisely the problem named open in [1]: Groskin [7] measures a Galerkin exponent but disclaims deriving it, and Śliwiński [15] establishes only a lower bound, so no prior work supplies the functional form given here. The novelty claimed is the law as a whole and an explicit, falsifiable program for identifying its mechanism; the underlying construction [1, 2], the Paley–Wiener strip mechanism [7], and the classical prolate and Mertens inputs are credited to their sources throughout.

8.2 Finite spectral reach and geometric zero capacity

The three budgets describe the accuracy of the selected first zero. The following result is orthogonal to those budgets: it records how far the retained frequency lattice extends and how many Riemann ordinates lie below that edge. It is placed after the assembled law because it is not a fourth error mechanism.

Proposition 6 (Exact retained-lattice edge and geometric capacity)**.** *Let $L = \ln\lambda^2$ and retain the Fourier–Mellin modes $|n| \le N$. Their frequencies are $\alpha_n = 2\pi n/L$. Hence the exact outer edge of the retained frequency lattice is*

$$\boxed{T_{\max}(N,\lambda^2) = \max_{|n|\le N} |\alpha_n| = \frac{2\pi N}{\ln\lambda^2} = \frac{N\pi}{\ln\lambda}.} \tag{20}$$

Let $N_\zeta(T)$ count the positive nontrivial zeros of ζ through height T, with multiplicity. Define the exact geometric capacity by

$$\boxed{K_{\max}^{\mathrm{geom}}(N,\lambda^2) := N_\zeta\big(T_{\max}(N,\lambda^2)\big)\,.} \tag{21}$$

As $N/\ln\lambda^2 \to \infty$,

$$K_{\max}^{\mathrm{geom}} = K_{\max}^{(0)} + S(T_{\max}) + O\!\left(\frac{\ln\lambda^2}{N}\right), \tag{22}$$

where

$$\boxed{K_{\max}^{(0)}(N,\lambda^2) = \frac{N}{\ln\lambda^2}\left[\ln\!\left(\frac{N}{\ln\lambda^2}\right) - 1\right] + \frac{7}{8}.} \tag{23}$$

Proof. The mode n has frequency $2\pi n/L$, so taking the maximum over $|n| \le N$ gives (20). Equation (21) is then a definition. The Riemann–von Mangoldt formula is

$$N_\zeta(T) = \frac{T}{2\pi}\ln\frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + S(T) + O(T^{-1}), \qquad S(T) = \pi^{-1}\arg\zeta\big(\tfrac{1}{2} + iT\big)\,,$$

with the usual limiting convention at an ordinate. Since $T_{\max}/(2\pi) = N/L$, substitution gives (23); moreover $T_{\max}^{-1} = L/(2\pi N)$, giving (22). □

Corollary 2 (Reach condition and smooth inverse capacity)**.** *The kth positive zeta ordinate lies inside the retained lattice edge exactly when*

$$\boxed{\gamma_k \le T_{\max} \quad\Longleftrightarrow\quad N \ge \frac{\gamma_k \ln\lambda^2}{2\pi}.} \tag{24}$$

If the oscillatory term is suppressed and the smooth equation $K_{\max}^{(0)}(N,\lambda^2) = k$ is inverted, then, with $a_k = k - 7/8 > 0$,

$$\boxed{N_{\mathrm{reach}}^{(0)}(k,\lambda^2) = \ln\lambda^2\,\frac{a_k}{W_0(a_k/e)},} \tag{25}$$

where W_0 is the principal Lambert W branch.

Proof. The first statement is (20) rewritten. For the second, put $y = N/\ln\lambda^2$. Then $y(\ln y - 1) = a_k$. Setting $z = \ln y - 1$ gives $ze^z = a_k/e$, so $z = W_0(a_k/e)$ and $y = a_k/W_0(a_k/e)$. □

The edge (20) is already implicit in the CCM definition $|t| \le N\pi/\ln\lambda$ and in the external carrier spectrum of [1, Sec. 5.6]; Groskin writes the same edge as ρN, with $\rho = 2\pi/\ln\lambda^2$, in a separate archimedean-tail problem [8]. The exact count (21) and the smooth formula (23) are different objects: the former is integer-valued and retains $S(T_{\max})$, while the latter is the smooth planning approximation. The operator does not use N_ζ, individual zeros, or any reference-zero table; the capacity is an analytic statement about the finite lattice after the operator has been defined.

To the best of the author's knowledge, the composition $N_\zeta(2\pi N/\ln\lambda^2)$ has not previously been stated as the geometric capacity of the finite CCM truncation. The qualification is essential: $T_{\max}$ is a retained-lattice edge, not a hard cutoff for zeros of the finite transform, and $K_{\max}$ does not prove accurate recovery, positivity, correct ordering, or one recovered zero per lattice interval.

8.3 Convergence criterion

Proposition 7 (Fixed-cutoff branch convergence)**.** *Fix $\lambda > 1$. Assume the spectral and zero-branch hypothesis*

$$(\mathrm{H}^*_\lambda): \quad \begin{cases} QW_\lambda \text{ has a simple, isolated, positive lowest eigenvalue } \varepsilon^\infty(\lambda), \\ \text{its normalized continuum eigenvector } \xi_\infty \text{ is even}, \\ \widehat{\xi}_\infty \text{ has a simple isolated positive zero } \nu_\infty(\lambda) \neq \gamma_1. \end{cases} \tag{26}$$

Let ξ_N be the phase-aligned finite-section ground states and let ν_N denote the zero of $\widehat{\xi}_N$ selected in a sufficiently small disc about ν_∞. Then:

(i) The lowest Rayleigh values converge monotonically to $\varepsilon^\infty(\lambda)$, while $\xi_N \to \xi_\infty$ in norm and $\widehat{\xi}_N \to \widehat{\xi}_\infty$ compact-uniformly.

(ii) For all sufficiently large N, the selected zero ν_N is unique in that disc and $\nu_N \to \nu_\infty(\lambda)$.

(iii) If the numerical root perturbation vanishes as $P \to \infty$, then

$$\lim_{\substack{N\to\infty \\ P\to\infty}} D(\lambda^2, N, P) = D_{\mathrm{Primes}}(\lambda^2) := -\log_{10}\frac{|\nu_\infty(\lambda) - \gamma_1|}{|\gamma_1|} < \infty.$$

Consequently, at this fixed cutoff the selected branch cannot converge to γ_1 merely by increasing N and P.

Proof. The Rayleigh-value statement is [1, Prop. 3.4]. Under the simple/isolated/even hypotheses in (26), [2, Thm. 6.1] gives convergence of the finite-section eigenvectors and compact-uniform convergence of their transforms. Choose a closed disc about ν_∞ containing no other zero of $\widehat{\xi}_\infty$. Since that zero is simple, Hurwitz's theorem (equivalently the local Rouché argument of Lemma 2) gives one zero ν_N in the disc for all sufficiently large N and $\nu_N \to \nu_\infty$. Vanishing numerical perturbation then gives the same limit as $P \to \infty$, and continuity of $-\log_{10}|\cdot|$ away from zero proves part (iii). □

Growing cutoffs. For a sequence $\lambda_j \to \infty$, convergence to γ_1 follows if the three separately defined errors vanish:

$$|\nu_{N_j}(\lambda_j) - \nu_\infty(\lambda_j)| \to 0, \qquad \varepsilon_{\mathrm{round}}(P_j) \to 0, \qquad |\nu_\infty(\lambda_j) - \gamma_1| \to 0.$$

This is a sufficient pathwise criterion from the triangle inequality. The budget choices $N_j \gtrsim (4/\pi)\lambda_j^2 \ln \lambda_j^2$ and $P_j \gtrsim K\lambda_j^2$, and the rate $D \sim K\lambda_j^2$, are the quantitative model tested in this paper; they do not follow from the qualitative fixed-cutoff theorem above. Conversely, ruling out convergence for an arbitrary family of cutoffs in a bounded interval would require a uniform separation $\inf_\lambda |\nu_\infty(\lambda) - \gamma_1| > 0$ on that interval, which is not claimed here.

Status of the hypothesis (H^*_λ)**.** We state (H^*_λ) explicitly because finite-cutoff Weil positivity is *not* known unconditionally for all λ: [1] is explicit that positivity of the continuum measure cannot be asserted in general, and positivity for all λ is precisely the open conjecture linking the construction to RH. What is unconditional is the small-support range: Yoshida proved the Weil hermitian form positive definite for test functions of sufficiently small support [16], Bombieri reproved and quantified this in his study of the truncated functional [17], Connes–Consani gave the operator-theoretic strengthening at the archimedean place via the Sonin space [18], and the ζ-cycles paper [19] tracks finite-λ positivity prime by prime through the first cutoffs. Suzuki [5, Thm. 1.4] also proves a positive, simple, even continuum ground state for sufficiently small aperture. Recent work isolates the pole block into one rank-one contribution in each parity sector and proposes a scalar Herglotz/Weyl reformulation of the remaining ordering problem [11, 12]. The Herglotz criterion depends on additional sector hypotheses that are not proved in the general arithmetic regime, so it is a structural reformulation rather than a proof of full even-simplicity. In that regime, proving the required odd-sector positivity would already imply RH through Yoshida's criterion. These results address only part of (H^*_λ); simplicity of the selected continuum-transform zero remains an explicit hypothesis. Beyond the rigorous small-aperture regime, positivity, simplicity, and evenness are supported numerically at every cutoff measured in this program ($\lambda^2 \le 200$, and $\lambda^2 \le 1000$ in [51]). The finite matrices $(\lambda^2, N) = (13, 120), (29, 160), (53, 200)$ additionally have independent Arb certificates that the lowest even-sector eigenvalue is simple and strictly below the lowest odd-sector eigenvalue [13]. Those certificates strengthen three finite data points but do not establish uniformity or the continuum limit. The noncoincidence $\nu_\infty(\lambda) \neq \gamma_1$ is likewise numerical and is what makes the fixed-cutoff limiting digit count finite.
The qualitative division of labor is now explicit. CCM Prop. 3.4 supplies monotone convergence only for the lowest Rayleigh values. Connes–van Suijlekom Thm. 6.1 supplies conditional eigenvector and compact-uniform transform convergence, and simplicity of the selected continuum zero supplies the zero branch. No monotonicity of root accuracy and no pointwise bound $D \le D_{\mathrm{Primes}}$ for every finite N is asserted: finite-N and cutoff displacements can partially cancel. The quantitative growing-cutoff rate inherits the separate rigor of D_{Primes}: the classical fourth-index prolate asymptotic is T1, while its transfer to the constrained CCM trial quotient is T3 and the continuum plunge remains an assumption.

Conceptually, along the selected branch the fixed-cutoff limit isolates the cutoff displacement once precision and Galerkin errors are driven to zero. Reaching γ_1 therefore requires a growing-cutoff path on which the continuum-transform zero itself approaches γ_1. The measured model $D \sim K\lambda^2$ quantifies the proposed cost along such a path — each additional digit demands a fixed increment $\ln 10/(4\pi)$ in λ^2 — but this quantitative rate is not supplied by the qualitative fixed-cutoff theorem.

8.4 Conditional Gram representation and divided-difference root transfer

The operational law above is empirical. This subsection gives a separate, conditional explanation of the finite eigenvalue-to-root conversion that the law currently records through a conversion gap. It is placed after the formula and its validation because the formula does not depend on the hypothesis used here.

$(\mathrm{H}_{\mathrm{RH}})$ *The Riemann Hypothesis. Ordinates γ are counted with multiplicity. Sim-*

plicity is required only where a simple-zero derivative or a one-to-one zero label is used.

The hypothesis is global: the sums below run over all nontrivial zeros, so a finite-height verification does not replace ($\mathrm{H_{RH}}$) at the depths considered here. We do not assign a numerical off-line remainder in this paper.

Proposition 8 (Finite dictionary and conditional even-sector Gram specialization)**.** *Let $V_N^+ \subset V_N = \operatorname{span}\{V_j : |j| \le N\}$ denote the real even sector, and let $f = \sum_{|j|\le N} u_j V_j \in V_N^+$ have $\|f\| = 1$. Groskin's finite Guinand–Weil dictionary associates to the coefficient vector of f an admissible entire band-limited function g_f and gives the unconditional identity*

$$QW_\lambda(f,f) = \sum_{z\in Z_\zeta^*} g_f(z), \qquad Z_\zeta^* = \{z \in \mathbb{C} : \tfrac{1}{2} + iz \text{ is a nontrivial zero of } \zeta\}, \tag{27}$$

with multiplicity [8, Thm. 2.5]. Writing $\rho = 2\pi/L$, Groskin's real-axis density and the present transform normalization give

$$g_f(T) = \frac{2\rho}{\pi}\sin^2(LT/2)\left|\sum_{|j|\le N}\frac{u_j}{T-\rho j}\right|^2 = \frac{4}{L}\sin^2(LT/2)\left|\sum_{|j|\le N}\frac{u_j}{T-\rho j}\right|^2 = |\widehat{f}(T)|^2.$$

Thus, under ($\mathrm{H_{RH}}$)*, equation* (27) *specializes to*

$$QW_\lambda(f,f) = \sum_{\gamma\in\pm\Gamma} |\widehat{f}(\gamma)|^2, \qquad \varepsilon_N^+(\lambda) = \min_{\substack{f\in V_N^+\\ \|f\|=1}} \sum_{\gamma\in\pm\Gamma} |\widehat{f}(\gamma)|^2. \tag{28}$$

The series is absolutely convergent for the induced finite-mode test functions. Under the finite-sector hypothesis that the simple global ground state lies in V_N^+, one has $\varepsilon_N^+(\lambda) = \varepsilon_N(\lambda)$.

Equation (27) is unconditional but is a sum over the actual complex zero set. The nonnegative modulus-square form (28) is the RH specialization. Equation (28) is a Christoffel-type extremal formulation of the even-sector eigenvalue depth. It unifies the eigenvalue-depth analogues of the two intrinsic channels: changing N changes the trial space, while changing λ^2 changes the exponential type and the exhausted prime content.

For the mixed pairing used below, support exhaustion and the classical Guinand–Weil formula applied to the symmetrized cross-correlation give, under ($\mathrm{H_{RH}}$),

$$\langle QW_\lambda f, h\rangle = \sum_{\gamma\in\pm\Gamma} \widehat{f}(\gamma)\,\overline{\widehat{h}(\gamma)} \tag{29}$$

for real $f, h \in V_N$; admissibility follows by the same finite-mode estimates as in [8, Lem. 2.2]. This bilinear specialization, rather than the even-sector minimum, is the identity used in the divided-difference argument. Root digits still require that separate eigenvalue-to-root transfer.

Proposition 9 (Divided-difference root transfer)**.** *Assume* ($\mathrm{H_{RH}}$)*. Let ξ be the unit ground vector of the finite form, with*

$$\widehat{\xi}_N(z) = 2L^{-1/2}\sin(zL/2)G_N(z), \qquad G_N(z) = \sum_{|j|\le N}\frac{\xi_j}{z-\alpha_j}, \qquad \alpha_j = \frac{2\pi j}{L}.$$

Let ν be a real zero of G_N and define $\eta_j = \xi_j/(\alpha_j - \nu)$. Then $\eta \in V_N$ and $\widehat{\eta}(z) = \widehat{\xi}_N(z)/(z-\nu)$. Testing the finite eigenvalue equation against η gives the exact identity

$$\sum_{\gamma\in\pm\Gamma}\frac{|\widehat{\xi}_N(\gamma)|^2}{\gamma-\nu} = \varepsilon_N S_N(\nu), \qquad S_N(\nu) = \sum_{|j|\le N}\frac{\xi_j^2}{\alpha_j-\nu}. \tag{30}$$

For the zero ν_k matched to the positive ordinate γ_k, set

$$B_{k,N} = \frac{\widehat{\xi}_N(\gamma_k)}{\gamma_k - \nu_k}, \qquad R_{k,N}(\nu) = \sum_{m \neq k} \frac{2\nu\,|\widehat{\xi}_N(\gamma_m)|^2}{\gamma_m^2 - \nu^2}.$$

Pairing the ordinates $\pm\gamma_m$ in (30) *yields*

$$\boxed{\gamma_k - \nu_k = \frac{\gamma_k + \nu_k}{2\nu_k}\,\frac{\varepsilon_N S_N(\nu_k) - R_{k,N}(\nu_k)}{|B_{k,N}|^2}.} \tag{31}$$

Proof. The partial-fraction identity

$$\frac{1}{(z-\alpha_j)(z-\nu)} = \frac{1}{\alpha_j - \nu}\left(\frac{1}{z-\alpha_j} - \frac{1}{z-\nu}\right)$$

shows that $G_N(z)/(z-\nu)$ is again a lattice sum; the apparent extra pole vanishes because $G_N(\nu) = 0$. Thus $\eta \in V_N$ and the matrix identity $\langle QW_\lambda \xi, \eta\rangle = \varepsilon_N \langle \xi, \eta\rangle$ becomes (30) by the bilinear specialization (29) following Proposition 8. Isolating the $\pm\gamma_k$ pair gives (31); no remainder is discarded. □

Define the finite eigenvalue depth $D_{\mathrm{eig},N} = -\log_{10}|\varepsilon_N|$ and

$$\widetilde{S}_{k,N} = S_N(\nu_k) - \frac{R_{k,N}(\nu_k)}{\varepsilon_N}.$$

Equation (31) gives the exact relative conversion gap

$$\boxed{D_{\mathrm{eig},N} - D_k = \log_{10}\!\left(|\widetilde{S}_{k,N}|\,\frac{\gamma_k + \nu_k}{2\gamma_k\nu_k|B_{k,N}|^2}\right).} \tag{32}$$

When $\nu_k \approx \gamma_k$ and $B_{k,N} \approx \widehat{\xi}_N'(\nu_k)$, this becomes

$$D_{\mathrm{eig},N} - D_k \approx \log_{10}\!\left(\frac{|\widetilde{S}_{k,N}|}{\gamma_k|\widehat{\xi}_N'(\nu_k)|^2}\right). \tag{33}$$

The remote term requires a truncated zero sum with a controlled tail; any published numerical use of it must state the truncation height, zero source, and tail estimate.

Relation to the doubling conjecture. The exact identity is linear in ε_N and contains two powers of a norming divided difference. It therefore identifies the conditional first-order mechanism for the total displacement $\nu_k(N) - \gamma_k$. Conjecture 1(iii), however, concerns $\nu_k(N) - \nu_k(\infty)$ at fixed cutoff. Promoting the measured doubled finite-basis exponent still requires subtracting the limiting identity and controlling

$$\varepsilon_N - \varepsilon^\infty, \qquad S_N - S_\infty, \qquad B_{k,N} - B_{k,\infty}, \qquad R_{k,N} - R_{k,\infty}.$$

Lemma 4 now supplies the Galerkin transfer from a sharp coefficient tail and a continuum gap to selected-zero convergence. What remains for the doubled exponent is the stronger subtracted control of the finite and limiting eigenvalue and norming terms displayed above.

Independent numerical check. On the independent implementation, the two sides of the first-order balance at $(\lambda^2, N, k) = (7, 40, 1)$ agree in ratio 0.999996. The exact conversion formula reproduces predicted/measured gaps 2.4689/2.4689 at $(7, 40)$, 2.69434/2.69435 at $(13, 80)$, and 2.78431/2.78432 at $(20, 144)$. Across $k \le 12$ at $\lambda^2 = 13, 20$, $\gamma_k S_N(\nu_k) \in [-1.07, -1.00]$, while the remote share decreases from 1.4% to 0.15% along the three increasingly saturated configurations. These are T3 measurements from an independent high-precision numerical implementation, not interval certificates. They are reported as supporting numerical evidence for the exact conditional identity and are not used to calibrate the operational law.

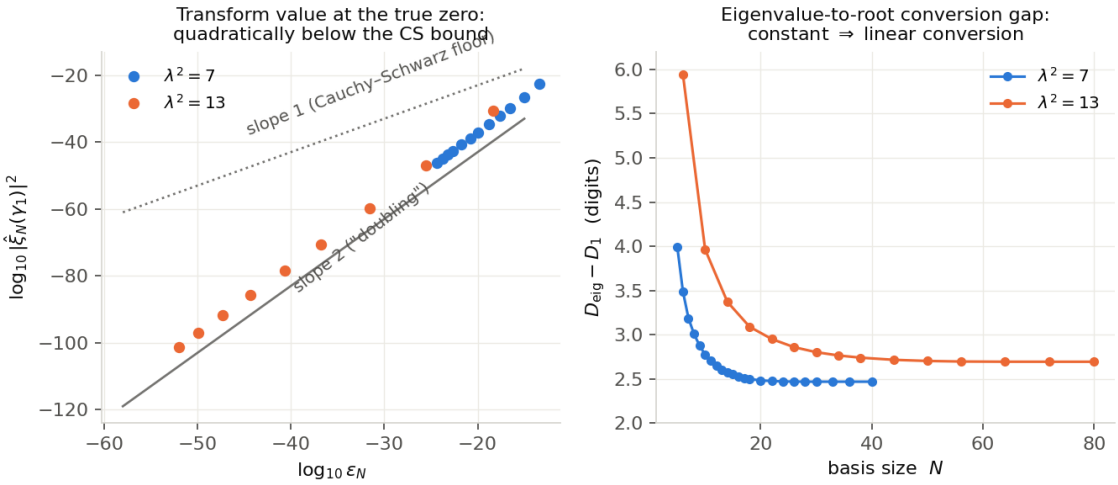


Figure 3: Independent evidence for the conditional root-transfer mechanism. The left panel shows the pre-saturation relation $|\widehat{\xi}_N(\gamma_1)|^2 \asymp \varepsilon_N^2$; it is evidence for the total-displacement mechanism, not by itself a proof of the strict fixed-cutoff doubling. The right panel shows the conversion gap approaching a cutoff-dependent plateau.

Status: the finite dictionary (27) *is T1 unconditionally. Its positive Gram specialization* (28)*, Proposition 9, and* (31) *are T1*$\,|\,(\mathrm{H_{RH}})$*. The observed near-constancy of* $\gamma_k S_N(\nu_k)$*, the remote-term estimates, and the numerical conversion gaps are T3. These conditional identities explain the conversion mechanism but do not enter the unconditional route from the CCM construction to RH.*

9 Discussion

Formula first. The empirical and analytic questions are deliberately separated. The first question is whether the specified inputs $(\lambda^2, N, P_{\mathrm{req}})$ produce an accurate digit prediction. Tables 2 and 3 answer that question directly: the fixed published submodel is accurate to about one digit on independent ramps through $\lambda^2 = 20$, remains useful with a residual of about six digits at the edge of the calibrated domain, and fails decisively when extrapolated unchanged to $\lambda^2 = 1000$. The second question is why the working formula has its observed form. The Gram, Galerkin, prolate, determinant, and prime-event programs address that second question; none is allowed to redefine the operational law without a better whole-cutoff validation result.

Relation to Groskin [7]. Groskin independently implemented the Connes–van Suijlekom matrix and measured a Galerkin convergence exponent $s(c)$ growing with the cutoff, proposing a Paley–Wiener analyticity-strip mechanism. The present work builds on that mechanism: Prop. 2(ii) gives the precise local-slope coordinate in which his $s(c)$ can be compared with our measured stretched-exponential. Establishing that the eigenvalue and root experiments sample

the same asymptotic tail remains part of the transfer problem. The candidate strip rate is the archimedean $\Gamma_{\mathbb{R}}$ value $\delta = \pi/4$. What is new here is the digit-space formulation, the Shannon coordinate, the calibrated closed form $q_\gamma(\lambda^2)$, and the embedding of the rate in the full three-budget minimum. A second independent reimplementation reproduced the fixed-form ramps at $\lambda^2 = 7, 13, 20$ and supplied the large-cutoff constraint in Section 5.5; this corroborates the law on its stated domain while preventing an unqualified extrapolation.

Groskin's revised analysis, extended to cutoff $\lambda^2 = 100$, is independent corroboration of this two-regime structure. He finds that no single smooth-convergence law fits his eigenvalue-space data and explicitly declines to assert a convergence rate, reporting instead an unresolved tension between a *drifting* effective exponent and an apparent *finite* convergence limit. Both halves have a common empirical interpretation here: Prop. 2(ii) explains how a stretched exponential produces the drift (its local exponent $2s_{\mathrm{loc}}(N) = aqN^q$ grows with N), while the prime ceiling of Section 6 supplies the finite limit $D_{\mathrm{Primes}}(\lambda^2)$. What looks like an irreconcilable choice within a single-rate framework is simply the pre-saturation ramp and the ceiling of the minimum law (2), seen in one dataset.

Relation to Śliwiński [15] and Suzuki [5]. Śliwiński proves an inverse-logarithmic lower bound on the *mean* spectral error over the whole spectrum (Thm. 3.1), a different regime from the first-zero convergence here, and consistent with it, since the mean is dominated by the unconverged high-index eigenvalues while $\nu_1 \to \gamma_1$ converges super-exponentially. Suzuki's screw-function analysis supplies the archimedean $\ln|z|$ multiplier and the prime-frequency support that underlie Sections 5.2–5.3.

Parity-resolved pole and prime structure. The pole block admits the exact decomposition

$$w_{02} = 2|C\rangle\langle C| - 2|S\rangle\langle S|,$$

so its even restriction is rank-one positive and its odd restriction is rank-one negative [11]. Groskin's dictionary gives the same even rank-one factorization and identifies a non-collapsing pole-neutral subspace [8, Cor. 2.7]. For the distinguished even vector C, Andrade proves a termwise non-positive prime contribution beyond an explicit finite-section threshold [10]. This is useful parity information, but it does not establish the sign required in Lemma 6 for the constrained CCM trial vector k_λ. No pole-free even-simplicity theorem is used here.

Origin of the leading rate: arithmetic, not archimedean. The prime ceiling is *modeled* by an archimedean object, the prolate eigenvalue of Section 6, yet the exponential rate is not carried by the archimedean part of the Weil form itself. Diagonalizing the screw-function split [5] at high precision, the archimedean-only localized form is $O(1)$-indefinite, with no exponentially small eigenvalue. Restricting it to the Sonin subspace, the orthogonal complement of the band-concentrated prolate modes, where Connes–Consani establish archimedean Weil positivity [18] (surveyed in [4]), restores positivity, but only as an $O(1)$ gap: the smallest Sonin eigenvalue is 2.26, 2.28, 2.57 at $\lambda^2 = 5, 8, 13$, essentially flat in λ^2, against a floor $K\lambda^2 = 27, 44, 71$. A direct diagonalization of the archimedean-only form at $\lambda^2 = 13$–50 finds the same structure from the eigenvector side: the lowest even mode is a delocalized Sonin-type state with an $O(1)$ eigenvalue, so the band-concentrated plunge eigenvector exists only for the full form. Archimedean positivity and the plunge are therefore *decoupled*: the place at infinity supplies an $O(1)$ spectral gap, while the exponentially small $\varepsilon^\infty \sim e^{-4\pi\lambda^2}$ is produced by the cancellation between the archimedean and prime parts of the full form. This locates one of the analytic steps behind the leading coefficient: a quantitative archimedean–prime cancellation estimate on the even ground eigenvector. The preceding constrained CCM-to-prolate quotient comparison is a separate open step, which is why the full CCM identification of K is T3.

Boundary spectral mass as a proof route. For a unit finite ground state let $w_0 = |\langle (2N+1)^{-1/2}\mathbf{1}, u_0\rangle|^2$. Across 56 independent configurations at $\lambda^2 \in \{7, 13, 20\}$, the ratio ε_N/w_0 remains between 0.398 and 1.762 while ε_N changes by more than 25 decimal orders. Because $\sum_j \xi_j$ is proportional to the boundary value of the finite Fourier state, this gives a Bonami–Karoui-type boundary-action interpretation of the depth [25]. Proposition 5 shows that the same w_0 is the exact spectral amplitude of every prime-power slope kink. The observed proportionality between w_0 and ε_N still requires a derived normalization before it carries theorem-level weight; at present it is one promising analytic attack on the mode law, not a replacement for it.

Next formula-selection experiment. The large-cutoff data do not yet choose among a corrected exponent, a changing onset coefficient, or a new coordinate. The next validation pass will compare five fixed candidate surfaces by leaving out entire cutoffs: the current q_γ model; the fitted linear relation for $1/q$; a saturated exponent; a logarithmically corrected pre-plunge coordinate; and a two-stage prediction $D_{\mathrm{eig},N} - C_{\mathrm{conv}}$. The selected replacement, if any, must improve the $\lambda^2 = 150, 200, 250, 500, 1000$ holdouts without degrading the existing $\lambda^2 \le 100$ surface.

Practical use. On the calibrated domain, the law is a parameter-selection guide: to target D^* digits, pick $\lambda^2 \gtrsim D^* \ln 10/(4\pi) = D^*/5.46$ so the ceiling clears D^*; use equation (6) to choose N so the ramp clears D^*; and choose P_{req} so the precision budget is nonbinding. Increasing any one parameter past the binding minimum yields nothing. At cutoffs beyond 100, the minimum and cutoff budgets remain useful, but the closed q_γ mode map must not be used for planning until the large-cutoff transition model is replaced or directly calibrated.

10 Rigor ledger

Table 4 collects the rigor status of every component of the law, together with the open item that would promote each to the next level. A fourth column types each open item: *named lemma* means the missing step is a precisely posed statement inside an existing classical theory (variational principle, Rouché, Szegő, Gevrey characterization), so that closing it is a matter of executing known machinery; *new analysis* means genuinely new estimates are required. The table preserves both categories as an explicit program for strengthening the paper.

Table 4: Rigor status of each component. T1: rigorous. T2: derived modulo explicitly named intermediate estimates. T3: derived in structure and validated against data, with a map not yet rigorous. T4: empirical. Open-item type: *named lemma* = precisely posed inside existing theory; *new analysis* = new estimates required.

Component	Status	Open item	Type
Minimum structure (Prop. 1), lower bound	**T1**	—	—
Minimum structure, matching upper bound	**T1** | (θ)	uniform quantitative noncancellation $E \ge \theta \max_i \varepsilon_i$	empirical check
$D_{\mathrm{Wprec}} = P_{\mathrm{req}} + O(1)$, slope 1	**T1** | stability	end-to-end backward error and conditioning audit	numerical analysis
Assembly-validity premise for empirical states	**T2**/validated	independent cutoff or closed-form check; residuals; structural and provenance gates	verified computation
Post-band archimedean tail and cutoff rule	**T1**	apply the explicit budget to every finite-T assembly	Groskin 3.2/3.3
Finite spectral edge $T_{\max} = 2\pi N/\ln\lambda^2$ (Prop. 6)	**T1**	—	—
Exact geometric capacity $K_{\max}^{\mathrm{geom}} = N_\zeta(T_{\max})$	**T1**	—	—
Smooth capacity $K_{\max}^{(0)}$ and remainder (eqs. (22)–(23))	**T1**	retain $S(T_{\max})$ for the exact integer count	—

Continued on next page

Table 4 (continued)

Component	Status	Open item	Type
Smooth inverse capacity (Cor. 2)	**T1**	oscillatory correction for exact inversion	Lambert W/zero counting
Finite prime-power path and simple-eigenvalue slope law (Prop. 5)	**T1** | simplicity	transfer the exact finite eigenvalue kink to transform-root displacement and the continuum limit	perturbation theory
Nyquist-cell zero-count parity (Prop. 3)	**T1**	quantitative cellwise counts require additional separation or slope bounds	real analysis
Stretched-exp rate unification (Prop. 2)	**T1**	—	—
General simple-zero displacement (Lem. 2)	**T1**	—	—
Tail-to-transform transfer (Lem. 4, Cor. 1)	**T1** | coefficient tail + continuum gap	prove Conj. 1(i) and the required continuum gap; no separate Galerkin hypothesis remains	new analysis
Lowest Rayleigh values at fixed cutoff	**T1**	—	CCM Prop. 3.4
Even-sector ordering at three certified finite matrices	**T1** (finite)	derive sectorwise projection constants and combine interval certificates with Liu's spectral-Galerkin lower-bound framework [38] to obtain a continuum sector gap	interval and spectral analysis
Reality of all transform zeros at those three matrices	**T1** (finite)	no extension beyond the certified configurations is claimed	certificate + CCM Thm. 5.10
Eigenstate and compact-uniform transform convergence	**T1** | (H_λ^*)	simple, isolated, even continuum ground state	CvS Thm. 6.1
Selected zero branch and fixed-cutoff digit limit (Prop. 7)	**T1** | (H_λ^*)	simple isolated transform zero; noncoincidence with γ_1	Hurwitz or Rouché
Growing-cutoff quantitative rate	**T3**	convergence of the continuum zero and the CCM/prolate transfer	new analysis
$n = 4$ prolate singular-value asymptotic (K, 9/2)	**T1**	—	—
D_{Primes} leading K: CCM floor transfer	**T3**	constrained h_0–h_4 quotient comparison and prime sign (Lem. 6)	new analysis
D_{Primes} leading K: reverse cutoff asymptotic	open	archimedean–prime cancellation	new analysis
D_{Primes} CCM floor slope 9/2	**T3**	transfer of the fourth-index Fuchs law; arithmetic inertness	new analysis
D_{Primes} matching offset (P_m, c_m)	**T4**	test $C_{\mathrm{conv}} \approx 3.03$ in the saturated limit; construct Z_λ and prove the uniform comparison	new analysis
Shannon-like coordinate $\lambda^2 \ln \lambda^2$	**T3**	prove the rate inputs and identify the exact time–band limiting operator	new analysis
Smooth Riemann–von Mangoldt density and $K_{\max}^{(0)}$	**T1**	—	phase counting/Stirling
Scaled CCM equilibrium realization	open	normalized determinant/logarithmic-potential limit and external field	new analysis
Archimedean matrix spectral distribution	**T2**	exact matrix class, logarithmic symbol, normalization, and arithmetic stability	new analysis
Common-edge interpretation (two scales)	**T3**	determinant-to-zero and spectral comparison; universality class; deep-edge constant	new analysis
Archimedean rate $\delta = \pi/4$	**T3**	finite-state Gamma-envelope realization, Conj. 1(ii), and doubling	new analysis
Closed exponent $q_\gamma = 2/(2\gamma + \ln\ln\lambda^2)$ on $7 \le \lambda^2 \le 100$	**T3**	calibrated-domain formula validated; unchanged large-cutoff extrapolation falsified	empirical model
Large-cutoff transition map	open	whole-cutoff comparison of exponent, onset, and logarithmically corrected coordinates	empirical and analytic
Onset constant $c = 4/\pi$ on the calibrated domain	**T3**	derive or replace the fitted argument coefficient; joint large-cutoff fit gives drift	new analysis
Finite Guinand–Weil dictionary on the real even sector (eq. (27))	**T1**	extension to mixed pairings supplied separately by the classical explicit formula	Groskin Thm. 2.5
Conditional positive Gram specialization (eq. (28))	**T1** | $(\mathrm{H}_{\mathrm{RH}})$	unconditional positivity would require control of off-line zeros	new analysis

Continued on next page

Table 4 (continued)

Component	Status	Open item	Type
Divided-difference transfer (Prop. 9)	**T1** | (H_{RH})	uniform limits of $S_N, B_{k,N}, R_{k,N}$ and the subtracted fixed-cutoff identity	new analysis
Finite conversion-gap formula (eq. (32))	**T1** | (H_{RH}) + **T3**	controlled remote-zero tail and saturated-limit asymptotics	new analysis

The minimum lower bound is unconditional; the equality direction holds under the quantitative noncancellation condition of Prop. 1, supported across the experimental program but not proved uniformly. T1 | (H^*_λ): the stated implication is rigorous conditional on the explicit spectral and zero-branch hypotheses in eq. (26). Yoshida–Bombieri establish the positivity component in the small-support range [16, 17]; the remaining hypotheses are stated separately rather than attributed to those results.

11 Conclusion

We have given a single law (2) for the convergence of the CCM zeta spectral triple. In it the minimum structure is forced; the measured prime ceiling has leading coefficient $4\pi/\ln 10$, matching the rigorous fourth-index prolate singular-value asymptotic. Its transfer through the constrained CCM trial state and the reverse inequality are open; and the mode budget, the convergence rate named open in [1], is measured as the logarithm of a stretched-exponential. The fixed operational map $q_\gamma = 2/(2\gamma + \ln\ln\lambda^2)$ with $c = 4/\pi$ predicts independent ramps through $\lambda^2 = 20$ to approximately one digit RMS and remains the validated operational formula on $7 \le \lambda^2 \le 100$. A new $\lambda^2 = 1000$ sweep rejects its unchanged extrapolation and places the effective exponent in the convention-dependent range 0.74–0.80; the data do not yet decide whether the correction belongs to the exponent, onset coefficient, or coordinate. Identifying the calibrated root-error exponent with a sharp eigenvector coefficient class remains open, and direct fits to finite archived eigenvectors show substantial N- and window-dependence that any continuum theorem must explain. The rigorous Riemann–von Mangoldt backbone and the measured convergence surface motivate the conjecture that the mode ramp and prime ceiling are two scales of one CCM spectral edge (Section 7). Under (H^*_λ), the selected finite-section zero converges to the fixed-cutoff continuum zero. Along a growing-cutoff path, vanishing numerical, Galerkin, and cutoff displacements is sufficient for convergence to γ_1; the explicit rate $D \sim (4\pi/\ln 10)\lambda^2$ remains the quantitative model tested here, not a consequence of the fixed-cutoff theorem. Each claim is stated with an explicit rigor level. The rigorous core (the minimum bound, the precision slope, the stretched-exponential reparametrization and local-slope identities of Prop. 2, and the general simple-zero displacement Lemma 2) stands on its own. Conditional on global RH, Proposition 9 adds an exact finite divided-difference formula for the eigenvalue-to-root conversion; it explains the measured conversion gap but is not part of an unconditional route to RH. The tail-to-transform mechanism is now derived in Lemma 4: the matrix cross-block bound, full Ritz-gap estimate, and uniform transform-evaluation bound are proved in Lemma 3. Its remaining analytic inputs are the sharp continuum coefficient tail and the continuum spectral gap, rather than a separate Galerkin hypothesis. Proposition 5 also converts Groskin's exact matrix-valued von Mangoldt jump into a one-sided eigenvalue and digit-depth slope law, with boundary spectral mass as the event amplitude. The variational bound of Prop. 4 supplies the unconditional trial inequality but does not identify its quotient with the prolate deficiency; and the remaining open items are typed in Table 4. Of these, the *named lemmas* (including the constrained-trial comparison and prime-sign hypothesis behind Lem. 6) are precisely posed inside classical theories and are natural next steps; the *new-analysis* items (the determinant/logarithmic-potential limit and spectral comparisons behind the common-edge conjecture, the coefficient-decay estimate for the $\Gamma_{\mathbb{R}}$-weighted form (Conjecture 1), the coefficient-class subordination behind q, the archimedean–prime cancellation needed for the reverse cutoff asymptotic, and positivity beyond

the Yoshida range) are where the construction's genuinely open mathematics lives. The paired eigenstate/root archive also isolates a more economical candidate for the finite-cutoff offset: the depth-to-relative-root conversion approaches about 3.03 digits over the largest precision-clear cutoffs sampled. If that constant persists after saturation, it favors the asymptotic logarithmic coefficient $P_m = 9/2$ and explains most of $c_m \approx 9.5$ without a growing conversion penalty. This is a new empirical constraint, not a proof of the target-transform comparison. Separately, Proposition 6 identifies the exact retained edge $T_{\max} = 2\pi N/\ln \lambda^2$, the exact integer capacity $K_{\max}^{\mathrm{geom}} = N_\zeta(T_{\max})$, and the smooth capacity (23), with the Lambert-W planning inverse (25). The detailed derivation is placed after the assembled law because reach is not a fourth budget. These quantities measure geometric availability before questions of convergence quality arise. For higher zeros, Proposition 9 remains an exact conditional diagnostic identity rather than an operational input.

12 Explicit non-claims

The per-component rigor of the law is the subject of Table 4 and is not repeated here. Beyond it, we are explicit about three boundaries.

- **No proof of the Riemann Hypothesis.** We characterize how the CCM/CvS approximation converges; the link from the construction to RH, continuum positivity of QW_λ for all λ, is the open conjecture of [1, 2, 4], and is neither assumed nor advanced here.
- **We do not establish that the construction reproduces the zeta zeros.** That the finite operator's zeros converge to the true ordinates as $\lambda^2 \to \infty$ is the construction's own open premise; we quantify the approach to γ_1 *under* that premise through the growing-cutoff criterion following Prop. 7, and do not prove the premise itself.
- **Scope.** The closed operational law is stated and tested for the first zero γ_1 of the Riemann zeta function. We make no claim about its constants for higher zeros, nor about extension to Dirichlet L-functions or other settings. The reach quantities $T_{\max}$ and $K_{\max}^{(0)}$ count geometric availability; they are not a higher-zero convergence or correct-ordering theorem. Proposition 9 applies conditionally to each fixed k, but it is an analytic identity rather than an operational higher-zero law. The fixed operational constants (q_γ, c) are validated on $7 \le \lambda^2 \le 100$. Historical reference values at $\lambda^2 = 150, 200$ remain excluded until their exact ramp provenance is recovered. The closed map has now been tested at $\lambda^2 = 1000$ and fails there: all fitting conventions give an effective exponent near 0.74–0.80, rather than $q_\gamma(1000) = 0.648$, and the unchanged formula overpredicts the measured ramp by more than 100 digits. We therefore make no claim that the calibrated q_γ or $c = 4/\pi$ persists at large cutoff. The minimum-budget structure, cutoff ceiling, and stretched-ramp family are separate claims and are not retracted by this failure.
- **No unconditional general finite-cutoff hypothesis.** Statements requiring the full spectral and zero-branch hypothesis (H_λ^*) are conditional. Yoshida and Bombieri establish its positivity component in the small-support range [16, 17], and Suzuki establishes a positive, simple, even continuum ground state for sufficiently small aperture [5, Thm. 1.4]; simplicity and noncoincidence of the selected transform zero remain explicit assumptions. We do not claim positivity of QW_λ for general λ.

Acknowledgments

The author is deeply grateful to Professor Alain Connes (Collège de France and IHÉS) for his encouragement, support, and ongoing correspondence. The author also thanks Professor Henri

Moscovici (The Ohio State University) and Professor Nigel Higson (The Pennsylvania State University) for correspondence and encouragement.

Statement on the Use of AI Tools

In accordance with current publication practices concerning authors' use of generative AI tools, the author discloses that Anthropic's Claude and OpenAI's Codex served as aids in preparing the exposition and the Rust implementation code, to specifications set by the author. AI assistance was used to improve prose formatting and readability, gather research material, propose candidate arguments, and validate calculations; all such output was reviewed by the author. The author originated, directed, and conducted every computational experiment presented here. The mathematical development was led by the author with assistance from AI. The author bears full responsibility for the contents of this manuscript.

Empirical testing and code availability

The results above rest on an extensive empirical program: over 5,000 high-precision CCM experiments across the cutoff, basis-size, and working-precision axes. Candidate rate laws were tested against the data and discarded when they failed [52]. Each budget was isolated by varying one parameter while keeping the other two nonbinding. The fixed published formula was then tested without per-cutoff refitting: an independent implementation gives RMS errors 0.47, 0.66, and 1.18 digits at $\lambda^2 = 7, 13, 20$, while a four-point $\lambda^2 = 1000$ sweep, independently reproduced by the Rust implementation, supplies the explicit out-of-domain failure in Table 3. The breadth of the resulting dataset ($\lambda^2 = 200$ at $N \geq 2400$ here and $\lambda^2 = 1000$ in the reproduction [51]) made those parameter-isolation and extrapolation tests possible.

A calibration state is admitted only with an artifact identity that records its numerical route and dependencies, a precision check, an archimedean cutoff or independently validated closed-form check, eigenpair residuals, structural symmetry tests, and reference-zero provenance. These gates are part of the reported empirical law: precision stability by itself is not treated as assembly correctness.

All tabulated results can be regenerated with the claim scripts supplied in the companion repository; each script writes its complete console log and structured measurement records. The paper source and reproduction scripts are available at `https://github.com/TeamXcelerator/ccm-convergence-rate`; the shared high-precision library (Xcelerator Toolkit) is at `https://github.com/TeamXcelerator/xcelerator-toolkit`. The integrity-bound high-precision reference-zero dataset is distributed with the toolkit. No proprietary input data are required. Both repositories are source-available in accordance with the licensing provided with each repository.

A Self-contained structural facts

This appendix records the two structural facts about $\widehat{\xi}$ used in the body, the closed form of the transform (Lemma 1) and entirety/reality of its zeros, in a form that depends only on elementary computation and on the self-contained proofs of [1]. Its purpose is to make explicit that no statement in this paper routes through the Carathéodory–Fejér machinery of [2], which we cite as the historical origin of the reality mechanism only.

A.1 The closed form of the transform, from scratch

Write $t = \ln u \in [-L/2, L/2]$, so that $V_n(u) = L^{-1/2} e^{2\pi i n(t+L/2)/L} = L^{-1/2}(-1)^n e^{2\pi i n t/L}$ and the Fourier–Mellin transform is the Fourier integral over the interval. For a single mode,

$$\begin{aligned}\widehat{V}_n(z) &= \int_{-L/2}^{L/2} e^{izt}\, V_n(t)\, dt = L^{-1/2}(-1)^n \int_{-L/2}^{L/2} e^{i(z+2\pi n/L)t}\, dt \\ &= L^{-1/2}(-1)^n\, \frac{2\sin\big((z+2\pi n/L)\, L/2\big)}{z+2\pi n/L}.\end{aligned}$$

Since $\sin(zL/2 + \pi n) = (-1)^n \sin(zL/2)$, the sign cancels:

$$\widehat{V}_n(z) = \frac{2L^{-1/2}\sin(zL/2)}{z+2\pi n/L}.$$

Summing over the eigenvector $\xi = \sum_{|n|\le N} \xi_n V_n$ and reindexing $j = -n$ (using $\xi_{-j} = \xi_j$ for the even eigenvector, or simply relabeling) gives

$$\widehat{\xi}_N(z) = 2L^{-1/2}\sin(zL/2) \sum_{|j|\le N} \frac{\xi_j}{z - 2\pi j/L},$$

the closed form of Lemma 1, in agreement with [1, Prop. 5.9]. The derivation is a one-line Fourier integral; no external input is used. Entirety is immediate: each apparent pole $z = 2\pi j/L$ is cancelled by the zero of $\sin(zL/2)$ there (the removable-pole computation in the proof of Lemma 1), so $\widehat{\xi}_N$ is entire of exponential type $L/2$.

A.2 Reality of the zeros: provenance

The statement that, in the critical case, all zeros of $\widehat{\xi}_N$ are real and coincide with the spectrum of the perturbed operator is [1, Thm. 5.10]. We record why its proof is self-contained there. The proof chain consists of: (a) [1, Lem. 5.2], the observation that the relevant matrices have the structure $\tau_{ij} = (b_i - b_j)/(i-j)$; (b) [1, Lem. 5.4], the elementary commutator identity $DT - TD = |\beta\rangle\langle\eta| - |\eta\rangle\langle\beta|$ for such matrices, with D the diagonal mode operator; and (c) the conclusion of [1, Thm. 5.10]: when ε_N is the smallest eigenvalue with even simple eigenvector, $T = QW_\lambda^N - \varepsilon_N \ge 0$, the perturbed operator is self-adjoint in the T-inner product, and the transform is proportional to a regularized characteristic determinant, $\det_{\mathrm{reg}}(D'' - z) = -i\lambda^{-iz}\widehat{\xi}(z)$, whose zeros are therefore real, being the characteristic polynomial of a self-adjoint matrix. Each of (a)–(c) is proven within [1] by direct computation and does not require the general Carathéodory–Fejér extension of [2]. The distinct qualitative passage from finite sections to the continuum eigenstate and its transform zeros in Prop. 7 does invoke [2, Thm. 6.1] under (H_λ^*).

A.3 The independent classical foundation

For completeness we record the pre-CvS lineage on which the truncated Weil form itself rests: Weil's positivity criterion; Yoshida's unconditional positivity for test functions of sufficiently small support [16]; Bombieri's study of the quadratic functional restricted to a finite interval, its minimizer in the L^2 unit ball, and the eigenvalues of its finite truncations [17]; the operator-theoretic strengthening of Yoshida's theorem at the archimedean place by Connes–Consani [18]; the prolate identification of the small eigenvectors in the ζ-cycles paper [19]; and Suzuki's screw-function framework [20, 21, 5], which unifies Yoshida, Bombieri, and the Connes–Consani(–Moscovici) results by means of continuous functions and de Branges-space methods. The construction studied in this paper can be founded entirely on this lineage together with [1].

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